

,, -PUT] ,, BRL ,, CODE
 , REVIS\$ #BJJJ

, DEVELOP\$, "U ! , SPONSOR%IP (!
 , BRL , AU?OR;Y (, NOR? , AM]ICA

, PUBLI%\$ BY
 , AM]ICAN , PR9T+ , H\SE =! , BL
 , L\ISVILLE1 , K5TUCKY
 , COPY"R #BJJJ 0, , BANA
 , ? MAT]IAL MAY 2 DUPLICAT\$ B N ALT]\$4

, 9 #A , VOLUME
 , BRL PAGES P#A-P#AF & #A-IB
 , PR9T PAGES ;I-;X & #A-C#GD P#A

, , BANA , , MEMB]S

, AM]ICAN , C\NCIL (! , BL 7, , ACB7
 , AM]ICAN , F.D,N =! , BL 7, , AFB7
 , AM]ICAN , PR9T+ , H\SE =! , BL 7, , APH7
 , ASSOCIAT\$, S]VICES =! , BL 7, , ASB7
 , ASSOCI,N = , \$UC,N & , REHABILIT,N (! , BL
 & , VISU,Y , IMPAIR\$ 7, , AER7
 , CALI=NIA , TRANSCRIB]S & , \$UCATORS (!
 , VISU,Y , H&ICAPP\$ 7, , CTEVH7
 , ! , CANADIAN , N,NAL , 9/ITUTE =! , BL
 7, , CNIB7
 , ! , CLOV]NOOK , CST]3 , OPPORTUNITIES =!
 , BL 7, , CCOB7
 , N,NAL , BRL , ASSOCI,N 7, , NBA7
 , N,NAL , BRL , PRESS 7, , NBP7
 , N,NAL , F\$],N (! , BL 7, , NFB7
 , N,NAL , LIBR>Y , S]VICE =! , BL &
 , PHYSIC,Y , H&ICAPP\$1 , LIBR>Y (, 3GRESS

7,,NLS_/, ,BPH7

,ROYAL ,NEW ,Z1L& ,F.D,N =! ,BL
7,,RNZFB71 ,ASSOCIATE ,MEMB]

P#C

,COPIES MAY 2 PUR*AS\$ F3

,AM]ICAN ,PR9T+ ,H\SE =! ,BL
,P4,04 ,BOX #FJHE
,L\ISVILLE1 , ,KY #DJBJF-JJHE
,TELEPH"03 #EJB-HIE-BDJE
, ,FAX3 #EJB-HII-BBGD
;,E-,MAIL3 _+INFO@APH.ORG_
,CATALOG ,NUMB]3 #E-BDDAJ-JJ

,,SPECIAL ,,SYMBOLS ,,US\$
,,9 ,,? ,,VOLUME

_ / PR9T SLA%

99 7#CE1 #CE7 REF];E 9DICATOR2 PLAC\$ AF
! ^WS 6: X REF]S & 2F ! NOTE

,A -PLETE LI/ (! , -PUT] ,BRL ,CODE
SYMBOLS MAY 2 F.D AT ! 5D (? VOLUME1
/>T+ ON PR9T PAGE #EC4

P#E

,,TRANSCRIB]'S , ,NOTES

,? VOLUME IS TRANSCRIB\$ US+ ,5GLI%
,BRL ,AM]ICAN ,EDI;N #AIID & ,NEME? ,BRL
,CODE = ,MA!MATICS & ,SCI;E ,NOT,N
#AIGB1 AC 6,BRL ,=MATS3 ,PR9CIPLES (
,PR9T 6,BRL ,TRANSCRIP;N #AIIG4
,9 ? VOLUME1 ": A 4CUS.N (, -PUT] ,BRL
,CODE SYMBOLS OR 9DICATORS 3TA9S !
SYMBOL BRLD ADJAC5T 6XS DOT NUMB]S1 !
USUAL 2G9 & 5D , -PUT] ,BRL ,CODE
9DICATORS REQUIR\$ = EMB\$D\$ NOT,N >E
OMITT\$4

```
,SEC;N                                     ,PAGE  
,,=E^W " " " " " " " " " " " " " " " " " " ;VII  
,,=E^W ,,6,,! #BJJJ ,,EDI;N " " " " " " " " ;VIII  
,,*ANGES ,,& ,,CL>IFIC,NS ,,6,,!  
    ,-PUT] ,,BRL ,,CODE ,,9 #BJJJ      ;IX  
  
#A.J ,,GUIDEL9ES ,,= ,,IDSTIFY+  
    ,-PUT] ,,NOT,N " " " " " " " " " " " " " "#A  
#A.A ,A ,NOTE " " " " " " " " " " " " " "#A  
    ,EXAMPLE #A.A.A ,NOTE 9 ,TEXT ""   "#A  
    ,EXAMPLE #A.A.B ,DOT ,MATRIX  
        ,PR9T " " " " " " " " " " " " " "#B  
#A.B ,PROGRAM ,L9ES " " " " " " " " " " " " "#B  
    ,EXAMPLE #A.B.A ,%ORT ,PROGRAM  
        ,L9ES ,IDSTIFI$ @,NUMBS ""     A#B  
    ,EXAMPLE #A.B.B ,SY/EMATIC  
        ,9D5T,N ( ,PROGRAM ,L9ES ""    "#C  
#A.C ,-;TS " " " " " " " " " " " " " A#C  
    ,EXAMPLE #A.C.A ,EXPLANATORY  
        ,-;TS " " " " " " " " " " " " " "#D  
    ,EXAMPLE #A.C.B ,EXPLANATORY  
        ,-;TS " " " " " " " " " " " " " "#D  
#A.D ,^WS " " " " " " " " " " " " " A#D P#G  
    ,EXAMPLE #A.D.A ,^W ,5CLOSE$ )9  
        ,ANGLE ,BRACKETS " " " " " " " " " " A#D  
#A.E ,HORIZONTAL ,SIGNS " " " " " " " " " " #E  
    ,EXAMPLE #A.E.A ,HORIZONTAL  
        ,SIGN ( ,-P>ISON " " " " " " " " " " #E  
    ,EXAMPLE #A.E.B ,HORIZONTAL  
        ,SIGN ( ,-P>ISON " " " " " " " " " " #E  
#A.F ,SUP]IMPOS$ SYMBOLS " " " " " " " " " " #E  
    ,EXAMPLE #A.F.A ,SUP]IMPOS$  
        ,SYMBOL " " " " " " " " " " " " " A#E  
#A.G ,^WS &/OR ,SYMBOLS " " " " " " " " " " #F  
    ,EXAMPLE #A.G.A ,^WS ,\ ( ,3TEXT   #F  
    ,EXAMPLE #A.G.B ,SYMBOL ,N  
        ,REPRES5T $ 9 ,PRIM>Y ,CODE ""  #F  
#B.J ,,,-PUT] ,,BRL ,,CODE  
    ,,SYMBOLS ,,&,USAGE " " " " " " " " " " #G  
#B.A ,!,BASIS (! ,,-PUT] ,BRL ,CODE   #G  
#B.B ,USE ( ,BRL ,SYMBOLS " " " " " " " " " " #G  
#B.C ,!,-PUT] ,BRL ,CODE--,A  
    ,SEP>ATE ,CODE " " " " " " " " " " " " A#G  
    ,EXAMPLE #B.C.A ,TE*NICAL-  
        ,APPE>+,TEXT 7,5GLI%,BRL7     #H  
    ,EXAMPLE #B.C.B ,TEXT ),EMBSD$,  
        ,-PUT] ,NOT,N " " " " " " " " " " " A#H  
    ,EXAMPLE #B.C.C ,TEXT ),EMBSD$,  
        ,WEB ,SITE ,A4RESS " " " " " " " " " " A#H  
    ,EXAMPLE #B.C.D ,,-PUT] ,PROGRAM   #I  
    ,EXAMPLE #B.C.E ,NONUSE ( T@1  
        9TO1 & @,PREC$+ ,,-PUT] ,BRL  
        ,CODE ,9DICATOR " " " " " " " " " " " #I  
    ,EXAMPLE #B.C.F ,NONUSE ( T@1
```

```

9T01 & 0,PRES$+ , -PUT] ,BRL
      ,CODE ,9DICATOR " " " " " " " " " " " " A#I
#C.J ,,=MAT " " " " " " " " " " " " #AA
#C.A ,-PUT] ,BRL ,CODE
      ,TRANSCRIP;NS " " " " " " " " " " " " #AA
#C.B ,TRANSCRIB]'S ,NOTE " " " " " " " " " " #AA
#C.C ,SYMBOLS " " " " " " " " " " " " #AA
#C.D ,BLANK ,L9ES " " " " " " " " " " " " A#AA
      ,EXAMPLE #C.D.A ,-PUT] ,NOT,N
      ,SET ,(F F ,BODY ( ,TEXT "" #AB
      ,EXAMPLE #C.D.B ,-PUT] ,NOT,N
      7,WEB ,SITE ,A4RESS7 ,SET
      ,(F ,F ,BODY ( ,TEXT " " " " " " " " " " #AC
#C.E ,STIRE ,-PUT] ,PROGRAM " " " " " " " " " " #AC
      ,EXAMPLE #C.E.A ,PROGRAM )
      ,IDSTIFY+ ,LABEL " " " " " " " " " " " " #AD
#C.F ,-PUT] ,NOT,N )9 ,TEXT " " " " " " " " " " #AG
      P#I
      ,EXAMPLE #C.F.A ,EMB$D$ ,-PUT]
      ,NOT,N " " " " " " " " " " " " A#AG
      ,EXAMPLE #C.F.B ,EMB$D$ ,-PUT]
      ,NOT,N ) ,RUNOV] " " " " " " " " " " " " A#AG
      ,EXAMPLE #C.F.C ,EMB$D$ ,-PUT]
      ,NOT,N 7,WEB ,SITE7 )
      ,RUNOV] " " " " " " " " " " " " A#AH
      ,EXAMPLE #C.F.D ,EMB$D$ ,-PUT]
      ,NOT,N ON ,NEW ,L9E " " " " " " " " " " " " A#AH
#D.J ,,UPP] ,,& ,,L[] ,,CASE
      ,,DESIGN,N " " " " " " " " " " " " #AI
#D.A ,-PUT] ,NOT,N_/ ,L[] ,CASE "" #AI
      ,EXAMPLE #D.A.A ,S+LE ,,BASIC
      ,PROGRAM ,TRANSCRIB$ )\T
      ,CASE ,9DICATORS " " " " " " " " " " " " A#AI
#D.B ,%IFT ,9DICATOR " " " " " " " " " " " " A#AI
#D.C ,CAPS ,LOCK ,9DICATOR " " " " " " " " " " #BJ
      ,EXAMPLE #D.C.A ,3SECUTIVE &
      ,S+LE ,UPP] ,CASE ,LRS "" #A#BJ
      ,EXAMPLE #D.C.B ,CASE
      ,DESIGN,N 9 ; ,E-MAIL
      ,A4RESSES " " " " " " " " " " " " A#BJ
#D.D ,CAPS ,REL1SE ,9DICATOR "" #B#BJ
      ,EXAMPLE #D.D.A ,CAPS ,LOCK &
      ,CAPS ,REL1SE ,9DICATORS "" B#BJ
#E.J ,,REPRESST,N ,( ,,3TROL ,, *S #BA
#E.A ,3TROL , *S " " " " " " " " " " " " #BA
      ,EXAMPLE #E.A.A 8,C>RIAGE
      ,RETURN0 ,REPRESST,N " " " " " " " " " " #BA
      ,EXAMPLE #E.A.B 8,C>RIAGE
      ,RETURN0 ,REPRESST,N " " " " " " " " " " #BA
#F.J ,,3T9U,N ,9DICATOR " " " " " " " " " " " " #BC
#F.A ,3T9U,N ,9DICATOR " " " " " " " " " " " " #BC
      ,EXAMPLE #F.A.A ,3T9U,N
      ,9DICATOR " " " " " " " " " " " " A#BC
#F.B ,RUNOV]S ( ,-PUT] ,NOT,N "" #A#BC
#F.C ,3T9U,N ,9DICATOR " " " " " " " " " " " " #BD
      ,EXAMPLE #F.C.A ,3T9U,N
      ,9DICATOR )9 ,EMB$D$
      ,-PUT] ,NOT,N " " " " " " " " " " " " A#BD
      ,EXAMPLE #F.C.B ,3T9U,N
      ,9DICATOR )9 ,EMB$D$ ,WEB
      ,SITE ,A4RESS " " " " " " " " " " " " A#BD
#G.J ,,9D5T$ ,,MAT]IAL " " " " " " " " " " " " #BE
#G.A ,LEVELS ( ,9D5T,N " " " " " " " " " " " " #BE
#G.B ,RUNOV]S " " " " " " " " " " " " #BE
      ,EXAMPLE #G.B.A ,9D5T$ ,MAT]IAL #BE
      ,EXAMPLE #G.B.B ,9D5T$ ,MAT]IAL #BF
#H.J ,,ISOLAT$ ,,L[]-CELL ,,SIGNS "" #BG
      P#AA
#H.A ,ISOLAT$ ,L[]-,CELL ,SIGNS "" #BG

```

```
,EXAMPLE #H.A.A ,ISOLAT$ ,L[]-  
    ,CELL ,SIGNS "##### #BG  
#I.J ,C.TA# ,SPACES "##### #BI  
#I.A ,C.TA# ,SPACES ,9DICATOR """" #BI  
    ,EXAMPLE #I.A.A ,C.TA# ,SPACES  
        ,9DICATOR "##### #BI  
#I.B ,C.TA# ,SPACES ) ,RUNOV]S """" #CJ  
    ,EXAMPLE #I.B.A ,C.TA# ,SPACES  
        ) ,3T9U,N ,9DICATOR "##### #CJ  
    ,EXAMPLE #I.B.B ,C.TA# ,SPACES  
        ,2G9N+ ON ,RUNOV] ,L9E """""" #CA  
#AJ.J ,EMPHASIS ,9DICATORS "##### #CC  
#AJ.A ,EMPHASIS ,9DICATORS "##### #CC  
    ,EXAMPLE #AJ.A.A ,EMPHASIS  
        ,9DICATORS "##### #CC  
    ,EXAMPLE #AJ.A.B ,EMPHASIS  
        ,9DICATORS "##### #CC  
#AJ.B ,TYPE ,/YLE "##### A#CC  
    ,EXAMPLE #AJ.B.A ,EMPHASIS  
        ,9DICATORS ) ,M ,?AN ,"O  
        ,TYPE ,/YLE "##### #CD  
    ,EXAMPLE #AJ.B.B ,EMPHASIS  
        ,9DICATORS ) ,M ,?AN ,"O  
        ,TYPE ,/YLE "##### #CE  
#AA.J ,,-PUT] ,BRL ,CODE  
    ,9DICATORS "##### #CG  
#AA.A ,,-PUT] ,BRL ,CODE ,9DICATORS #CG  
#AA.B ,CAPS ,LOCK ,9DICATOR """""" #CG  
    ,EXAMPLE #AA.B.A ,,-PUT] ,BRL  
        ,CODE ,9DICATORS "##### A#CG  
    ,EXAMPLE #AA.B.B ,,-PUT] ,BRL  
        ,CODE ,9DICATORS "##### A#CG  
    ,EXAMPLE #AA.B.C ,AVOID+  
        ,FREQU5T ,SWIT*ES ,2T  
        ,,-PUT] ,BRL ,CODE &  
        ,PRIM>Y ,CODE "##### #CH  
    ,EXAMPLE #AA.B.D ,USE ( ,,-PUT]  
        ,BRL ,CODE ,9DICATORS  
        6,CL>IFY ,AMBIGUITIES """""" A#CH  
#AB.J ,NEME? ,CODE ,9DICATORS """" #CI  
#AB.A ,NEME? ,CODE "##### #CI  
    ,EXAMPLE #AB.A.A ,NEME? ,CODE  
        ,9DICATORS "##### #DJ  
    ,EXAMPLE #AB.A.B ,NEME? ,CODE  
        ,9DICATORS ) ,HORIZONTAL  
        ,FRAC;N ,L9E """""""""" #DA  
#AB.B ,NEME? ,CODE ,9DICATORS """" A#DA  
  
P#AC  
    ,EXAMPLE #AB.B.A ,NEME? ,CODE  
        ,9DICATORS }9 ,,-PUT] ,BRL  
        ,CODE "##### A#DA  
#AC.J ,,%APE ,9DICATORS "##### #DC  
#AC.A ,%APE ,9DICATORS "##### #DC  
    ,EXAMPLE #AC.A.A ,%APE  
        ,9DICATORS "##### #DC  
    ,EXAMPLE #AC.A.B ,%APE  
        ,9DICATORS "##### #DC  
#AC.B ,,-PUT] ,KEYBO>D "##### A#DC  
    ,EXAMPLE #AC.B.A ,DEPICT$ ,KEY #DD  
#AC.C ,TRANSCRIB]S ( ,NEME? ,CODE #DD  
    ,EXAMPLE #AC.C.A ,USE ( ,NEME?  
        ,CODE )9 ,%APE ,9DICATORS "" #DD  
#AC.D ,A ,TRANSCRIB]'S ,NOTE """" A#DD  
#AD.J ,TRANSCRIB]'S ,OP;N  
    ,SYMBOLS "##### #DE  
#AD.A ,TRANSCRIB]'S ,OP;N ,SYMBOL #DE  
    ,EXAMPLE #AD.A.A ,TRANSCRIB]'S  
        ,OP;N ,SYMBOL ,REPRES5T+
```

```

,, -PUT] ,, BRL ,, CODE      ; I-; VII
    , REVIS$ #BJJJ
    ,, =E^W

```

,Z \R SOCIETY'S DEP5D;E ON -PUT+
TE*NOLOGY GR[S1 -PUT]S 2COME 9CR1S+LY
IMPORTANT 6! BL & VISU,Y IMPAIR\$4 ,^? :O
R1D BRL 9 ORD] 6LE>N AB -PUT]S &
6-MUNICATE) !M H AN URG5T NE\$ =A BRL
-PUT] CODE T IS UNAMBIGU\S1 & T MAKES
POSSI# ! FAI?;L TRANSCRIP;N 96BRL (
-PUT]-RELAT\$ TEXT 9 PR9T4 ,9 #AIGB1
,,BANA RECOGNIZ\$? NE\$ 0ADOPT+ ,!

,PROVI.NAL ,BRL ,CODE = , -PUT] ,NOT,N4
 ,X 0 A PRELIM9>Y CODE T LEFT _M PRO#MS
 UNSOLV\$1 B XS ADOP;N ALL[\$ BRL R1D]S &
 BRL TRANSCRIB]S 6A3UMULATE A BODY (
 EXP]I;E ON : 6BASE A M E6ICI5T CODE4 ,!
 CODE PRES5T\$ & EXPLICAT\$ 9 ? VOLUME HAS
 BE5 MADE POSSI# 0T EXP]I;E4

,! , -PUT] ,BRL ,CODE UTILIZES ! SIXTY-
 F\R -B9,NS (DOTS T C 2 =M\$ 9 ! BRL
 CELL1 & ASSIGNS 6^! DOT PATT]NS ! #A
 SAME M1N+S ASSIGN\$ 6!M 0,NOR? A;VII
 ,AM]ICAN MANUFACTUR]S (-PUT] BRL
 EQUIP;T4 ,H["E1 AN E6ECTIVE BRL CODE M/
 2 CAPA# (REPRES5T+ ! #ABH "*"S 9 !
 ,,ASCII CODE1 & M/ PROVIDE A FEW A4I;NAL
 SYMBOLS T TRANSCRIB]S NE\$ 9 ORD] 6A*IEVE
 FAI?;L BRL TRANSCRIP;NS4 ,6MEET ^!
 REQUIRE;TS1 TWO M1N+S M/ 2 ASSIGN\$ 6_M (
 ! SIXTY-F\R DOT PATT]NS1 &A WAY (
 9DICAT+ ! M1N+ T IS 9T5D\$ M/ 2 PROVID\$4
 ,9 ! , -PUT] ,BRL ,CODE1 ! DI6]5TI,N 2T !
 TWO M1N+S ASSIGN\$ 6A DOT PATT]N IS
 A3OMPLI%\$ 0M1NS (A PREFIX 3SI/+ (DOTS
 #DEF4

,! , -PUT] ,BRL ,CODE , -MITTEE HAS /UDI\$
 A L>GE & DIV]SE SAMPLE (CURR5T -PUT]
 MAT]IALS1 & HAS "W\$ DILIG5TLY 6MAKE !
 , -PUT] ,BRL ,CODE A R1LI/IC CODE1 CAPA#
 (UNAMBIGU\S REPRES5T,N (CURR5T -PUT]
 NOT,N1 B FLEXI# 5 6RESPOND 6*ANG+ &
 EXP&+ NE\$S4 ,FE\$BACK F BRL R1D]S1
 TRANSCRIB]S1 & PUBLI%]S HAS BE5
 SOLICIT\$1 & IS REFLECT\$ 9 ? EDI;N4

B;VII
 ,,BANA & ? , -MITTEE RECOGNIZE T ! CODE
 DOES N A4RESS IMPORTANT ISSUES1 S* Z !
 TRANSCRIP;N (FL[*>TS & GRAPHICS4
 ,H["E1 WE >E PL1S\$ 6PUBLI% X 9 XS PRES5T
 =M1 & TRU/ T A FUTURE REVI.N W D1L) S*
 ISSUES4 ,WE WELCOME SU7ES;NS & -;TS F !
 FIELD4

,,BANA WI%ES 6?ANK ,DR4 ,T4 ,V4 ,CRANM]
 7,*AIRMAN (! , -PUT] ,BRL ,CODE
 , -MITTEE71 & ,DR4 ,EM]SON ,F\LKE1 ,MRS4
 ,PRISCILLA ,H>RIS1 ,MISS ,DONNA ,PA/ORE1
 ,DR4 ,S&RA ,RUCONI*1 ,DR4 ,LAWR;E
 ,SCA451 & ,MR4 ,JOSEPH ,SULLIVAN
 7-MITTEE MEMB]S71 = _! "W4

#C
 ,,=E^W ,,6,,! #BJJJ ;VIII
 ,,EDI;N

,S9CE ! #AIHG PUBLIC,N (! , -PUT] ,BRL
 ,CODE1 ! USE (! 9T]NET HAS 2COME
 -MONPLACE4 ,? EDI;N 3TA9S "S NEW1
 9T]NET-RELAT\$ EXAMPLES : DID N EXI/ 9
 #AIHG4

,! NUMB]+ (RULES & EXAMPLES HAS *ANG\$
 6REFLECT ! ,,BANA DOCU;T DEVELOP\$ 9
 #AIIH1 ,,SPECIFIC,NS ,,= ,,! ,,PRODUC;N
 ,,(,,BRL ,,& ,,PR9T ,,PUBLIC,NS ,,= ,,!
 ,,BRL ,,AU?OR;Y ,,(,,NOR? ,,AM]ICA4
 ,EA* SEC;N /&S AL"O1 MAK+ FUTURE A4I;NS
 EASI]4

,SEC;N #C.F1 ! RULE T D1LS)! PLACE;T (
 EMB\$D\$ -PUT] NOT,N1 HAS BE5 CL>IFI\$
 6A4RESS ! ISSUE (": 62G9 A PASSAGE (
 EMB\$D\$ -PUT] NOT,N : REQUIRES M ?AN "O
 BRL L9E4

,! NEW EXAMPLES &! CL>IFI\$ RULE >E LI/\$
 ON PAGE ;IX1 & AN 9DEX HAS BE5 A4\$ 6!
 CODE = EASI] REF];E4 A;VIII

,,BANA ,,-PUT] ,,TE*NICAL
 ,,-MITTEE

,EILE5 ,CURRAN 7,3SULTANT7
 ,,\$W>D ,GODFREY
 ,MELISSA ,HIR%SON 7,3SULTANT7
 ,JOHN ,JACKSON
 ,DR4 ,S&RA ,RUCONI*
 ,JOSEPH ,SULLIVAN
 ,BETTY ,T1*MAN 7,*AIR7

,,*ANGES ,,& ,,CL>IFIC,NS #E
 ,,6,,! ,,-PUT] ,,BRL ,,CODE ;IX
 ,,9 #BJJJ

,EXAMPLE #A.C.A;D *ANG\$
 ,EXAMPLE #A.G.B NEW

,EXAMPLE #B.C.C NEW
 ,EXAMPLE #B.C.F NEW

,EXAMPLE #C.D.B NEW
 ,EXAMPLE #C.E.A NEW
 ,SEC;N #C.F CL>IFI\$ RULE
 ,EXAMPLE #C.F.C NEW
 ,EXAMPLE #C.F.D *ANG\$

,EXAMPLE #D.C.B NEW

,EXAMPLE #F.C.B NEW

,EXAMPLE #AJ.B.B NEW

,EXAMPLE #AA.B.B NEW

,EXAMPLE #AC.A.B NEW
 ..#A.J ,,GUIDEL9ES ,,= ;X-#A
 ,,ID5TIFY+ ,,-PUT] ..,NOT,N3 ,! ,,-PUT]


```
,BRL ,CODE IS 5T]$ :5 PRECISE REPRES5T,N
IS ESS5TIAL4 ,?US1 MAT]IAL S* Z
PROGRAMS1 PROGRAM L9ES1 -PUT] -M&S1 &
FILE"NS %D 2 TRANSCRIB$ 9 , -PUT] ,BRL
,CODE4 ,! FOLL[+ GUIDEL9ES MAY 2 HELP;L
9 ! ID5TIFIC,N ( -PUT] NOT,N4 ,2C
LANGUAGES & SITU,NS ! TRANSCRIB] MAY
5C.T] W D\BT.S V>Y WIDELY1 ^! GUIDEL9ES
>E (F]$ M]ELY Z SU7ES;NS & N Z ALL-
9CLUSIVE1 B9D+ RULES4
```

```
#A.A ,A NOTE AT ! 2G9N+ (A "W MAY
DESCRIBE H[ -PUT] NOT,N IS REPRES5T$4
,EV5 IF NO S* NOTE EXI/S1 /YLI/IC
PRACTICES 7E4G41 TYPE /YLES DI6]5T F
SURR.D+ TEXT--DOT MATRIX PR9T1 = 9/.E7
MAY 9DICATE ! PRES;E ( -PUT] NOT,N4
,EXAMPLE #A.A.A3 ,NOTE 9 ,TEXT
```

```
,YR 3SOLE 4PLAYS B0? :AT Y TYPE &!
-PUT] RESPONSE4 ,? BOOK USES ! FOLL[+
3V5;N 6DI6]5Tiate 2T \TPUT F ! -PUT] &
DATA TYP$ AT ! KEYBO>D3 ,ALL OP]ATOR
,SEC;N #A.J-A.A #G
9PUT IS "UL9$1 :ILE -PUT]-G5]AT$ A#A
\TPUT IS N4
,"H IS AN EXAMPLE3
```

```
_A> *_>DIR *._>COM_<<CR>_/
_A: _>FORMAT.COM:MBASIC.COM:PIP._&
COM:ED.COM
-----#B
,EXAMPLE #A.A.B3 ,DOT ,MATRIX
,PR9T
```

DOS READY

```
#A.B ,PROGRAM L9ES >E G5],Y
RELATIVELY %ORT & "S"TS 2G9 ) NUMB]S4
,(T5 ^! L9ES >E 9D5T$ 9 A SY/EMATIC WAY
: 4T+UI%ES !M F MA!MATICS OR REGUL>
TEXT4
```

```
,EXAMPLE #A.B.A3 ,%ORT A#B
,PROGRAM ,L9ES ,ID5TIFI$ 0,NUMB]S
```

```
10 PRINT "HOW OLD ARE YOU?"
20 INPUT A
30 PRINT "THANK YOU!"
40 PRINT "YOU CAN RETIRE"
50 PRINT "IN" _; 65-A _; "YEARS."
60 END
```

```
-----#C
,EXAMPLE #A.B.B3 ,SY/EMATIC
,9D5T,N ( ,PROGRAM ,L9ES
```

```
_ *FOR/_ _I := _1 _*TO/_ _&
_NO_OF_LINES _*DO/_
_*BEGIN/_
_READLN(_LINE);
_*IF/_ _LENGTH(_LINE) < _LIMIT _&
_*THEN/_
_SHORT_LINES := _SHORT_LINES + _1
_*ELSE/_
```

```

_LONG_LINES := _LONG_LINES + _1
_*END_/;

,SEC;N #A.A-A.B #I
#A.C , -;TS "S"TS EXPLA9 :AT IS A#C
HAPP5+ 9 A PROGRAM4 , ^! -;TS >E
FREQU5TLY 4T+UI%$ F SURR.D+ MAT]IAL
0"*I/IC SYMBOLS OR ^WS ,7E4G41 _+/* _&
... */ , (* ... *) , _>REM, _>OUTPUT_:1
ETC47'4
-----#D
,EXAMPLE #A.C.A3 ,EXPLANATORY
, -;TS 7#E-JJ7

A4 _+/* _THIS IS A COMMENT IN _C */_/:
B4 _+(* _THIS IS A COMMENT IN _&
_PASCAL *)_:
C4 _+[_ _SO IS THIS _]_:
D4 _+110 REM THIS IS A COMMENT IN _&
BASIC_:

,NOTE3 , ^! PROGRAM L9ES >E TR1T$ Z
EMB$D$ 9 ! EXAMPLES 2C (! ID5TIFY+ LRS4
,9 AN ACTUAL PROGRAM LI/+ !Y WD 2 9
4PLAY$ NOT,N1 &! 9DICATORS WD N 2 US$4
,EXAMPLE #A.C.B3 ,EXPLANATORY
, -;TS

_[ _LONGEST POSSIBLE REPRESENTATION _]
#A.D , ^WS MAY 2 5CLOS$ )9 A#D
QUOT,N M>KS OR ANGLE BRACKETS4
,EXAMPLE #A.D.A3 , ^W ,5CLOS$ )9
,ANGLE ,BRACKETS

<FILE>
-----#E
#A.E ,HORIZONTAL SIGNS ( -P>ISON
7E4G41 _+<=, ==, :=_:7 MAY 2 PRES5T4
,EXAMPLE #A.E.A3 ,HORIZONTAL ,SIGN
( , -P>ISON

IF (INWORD == 0)

,EXAMPLE #A.E.B3 ,HORIZONTAL ,SIGN
( , -P>ISON

(EV _^= EV_NULL)=>

#A.F ,SUP]IMPOS$ SYMBOLS 7E4G41 A
CIRCLE )A SLA% 9SIDE7 MAY APPE>4

,SEC;N #A.C-A.F #AA
,EXAMPLE #A.F.A3 ,SUP]IMPOS$ A#E
,SYMBOL

,TO _>OVERLAY_: "O " * ON TOP ( ANO!R =!
PURPOSE ( HAV+ TWO OR M " *S AT ! SAME " *
POSI;N AT PR9T "T1 E4G41 _$2H1V_:_:3

,NOTE3 , ! %APE 9DICATORS 5CLOSE #B;H#A;V
6DESCRIBE ! %APE 3SI/+ ( TWO HORIZONTAL
L9ES & "O V]TICAL L9E4
-----#F
#A.G , ^WS &_/OR SYMBOLS MAY N MAKE
S5SE 9 ! 3TEXT (! PRIM>Y CODE 2+ US$4
,EXAMPLE #A.G.A3 , ^WS , \ ( , 3TEXT

```

10 LPRINT "HELLO"

,EXAMPLE #A.G.B3 ,SYMBOL ,N
 ,REPRESENT\$ 9 ,PRIM>Y ,CODE 7#E-JJ7

BRUCE__R.__FORTNUM@MAGIC.CA

,NOTE3 ,? ;,E-MAIL A4RESS 3TA9S "USCORE
 "*S 7#DEF1 #DEF7 : >E N REPRESENT\$ 9
 ,5GLI% ,BRL4
 ..#B.J ,,-PUT] ,BRL ,CODE #G
 ,,SYMBOLS ,,& ,,,USAGE
 #B.A ,! ,,-PUT] ,BRL ,CODE IS BAS\$ ON
 ,,ASCII 7,AM]ICAN ,/&>D ,CODE = ,9=M,N
 ,9T]*ANGE71 A CODE US\$ 0VIRTU,Y ALL
 -PUT]-RELAT\$ BRL DEVICES 4TRIBUT\$ 9
 ,NOR? ,AM]ICA4 ,PR9TA# SYMBOLS F O!R
 CODES 7E4G41 ,EBCDIC & ,BAUDOT7 C AL 2
 REPRESENT\$ US+ ,,-PUT] ,BRL ,CODE
 SYMBOLOGY4 ,! ,,-PUT] ,BRL ,CODE MAKES
 -MUNIC,N) BRL DEVICES POSSI# :ILE
 SIMULTANE\SLY PROVID+ =! PRECISE
 TRANSCRIP;N (-PUT]-RELAT\$ MAT]IALS 9
 BRL4 ,A TA# (,,-PUT] ,BRL ,CODE SYMBOLS
 APPE>S AT ! 5D (? CODE4

#B.B ,AL? 9 C]TA9 9/.ES _M BRL
 DEVICES USE "O BRL SYMBOL 6REPRESENT TWO
 DI6]5T PR9T "*S 7E4G41 UPP] CASE ;,A &
 L[] CASE ;A >E B0? BRLD Z DOT #A71 !
 ,,-PUT] ,BRL ,CODE AVOIDS ? AMBIGU;Y4
 ,/&>D ALPHABETIC "*S 7;A-;Z7 >E US\$ =
 L[] CASE LRS4 ,! SAME SYMBOLS1 PREC\$\$
 0DOTS #DEF1 >E US\$ 6REPRESENT UPP] CASE
 LRS4 ,BRL SYMBOLS US\$ 9 ? DUAL MANN]

,SEC;N #A.F-B.B #AC
 9CLUDE S+LE UPP] CASE ;,A & L[] A#G
 CASE ;A LRS1 AT SIGN _+(@) AND GRAVE _&
 ACCENT (_@), UP ARROW/CIRCUMFLEX (^) _&
 AND TILDE (_^), LEFT BRACKET ([]) AND _&
 LEFT BRACE (_[]), RIGHT BRACKET (]) _&
 AND RIGHT BRACE (_]), AND BACKSLASH _&
 (\) AND VERTICAL BAR (_\)_:4 ,! "USCORE
 SYMBOL 7_+__:7 HAS A SEP>ATE ID5T;Y4

#B.C ,! ,,-PUT] ,BRL ,CODE IS A
 SEP>ATE1 SECOND>Y CODE1 : MAY 2 US\$ 9
 3JUNC;N)! PRIM>Y CODE 9 : ! TEXT IS
 TRANSCRIB\$--:E!R T CODE 2 ,5GLI% ,BRL1
 ,BRL ,=MATS1 ,NEME?1 OR ,BRL ,MUSIC4
 ,MAT]IAL S* Z NUMB]S1 A2REVI,NS1
 ACRONYMS1 3TRAC;NS1 & PUNCTU,N M>KS %D 2
 TRANSCRIB\$ 9 ? PRIM>Y CODE UN.S " ! IS GD
 R1SON 65T] ,,-PUT] ,BRL ,CODE4 ,?US1
 TE*NICAL-APPE>+ A2REVI,NS & ACRONYMS S*
 Z 8,,IBM0 & 8,,ASCII0 03URR+)9 AN
 ,5GLI% ,BRL TRANSCRIP;N %D 2 TRANSCRIB\$
 9 ,5GLI% ,BRL4 ,,-PUT] ,BRL ,CODE IS 5T]\$
 :5 PRECISE REPRESENT,N IS ESS5TIAL2 = ?
 R1SON1 ,,-PUT] ,BRL ,CODE MAT]IAL IS
 TRANSCRIB\$ "*" = "*" 7I4E41 NO 3TRAC;NS >E
 US\$71 &! 3TRAC;NS TO1 9TO1 & 0MAY N 2
 US\$ 2F ANY ,,-PUT] ,BRL ,CODE B#G
 9Dicator4 ,,-PUT] ,BRL ,CODE NUMB]S >E
 TRANSCRIB\$ 9 ! L[] TWO-?IRDS (! CELL &
 >E N PREC\$\$ 0NUMB] SIGNS4 ,PROGRAMS &
 O!R -PUT] NOT,N SET (F F ! BODY (! TEXT
 & -PUT] 9PUT 9 : E LR M/ 2 TYP\$
 PRECISELY 9 ! MANN] %[N >E EXAMPLES (
 MAT]IAL : %D 2 TRANSCRIB\$ 9 ,,-PUT] ,BRL
 ,CODE4 ,:5 9 D\BT1 ,,-PUT] ,BRL ,CODE %D

```

2 US$4 ,LIKEWISE1 ! TRANSCRIB] M/ AL
JUDGE OR SEEK ADVICE Z 6! SIGNIFIC.E (
V]TICAL & HORIZONTAL SPAC+ )9 ! TEXT4
,:5 9 D\BT1 ASSUME SPAC+ IS SIGNIFICANT4
-----#H
,EXAMPLE #B.C.A3 ,TE*NICAL-,APPE>+
,TEXT 7,5GLI% ,BRL7

```

```

, " ! 7 _M -PLA9TS AB ! SIZE ( , ,BCDIC1
& , ,IBM 9TRODUC$ ,EXT5D$ , ,BCDIC )
,SY/EM_/#CFJ4

```

```

,SEC;N #B.B-B.C #AE
,EXAMPLE #B.C.B3 ,TEXT ) A#H
,EMB$D$ , -PUT] ,NOT,N 7,5GLI% ,BRL &
, -PUT] ,BRL ,CODES7

```

```

,! TE/ CRIT]ION -P>ES _+J_: 6! T]M VALUE
7#B7 _+3000_: 7_+!=_: TE/S = 9EQUAL;Y74

```

```

,NOTE3 ,! 7#B7 REF]S 6A SECOND ITEM (
PREVI\SLY 4CUSS$ MAT]IAL4
,EXAMPLE #B.C.C3 ,TEXT ) ,EMB$D$
,WEB ,SITE ,A4RESS 7,5GLI% ,BRL & , -PUT]
,BRL ,CODES7 7#E-JJ7

```

```

.,5G9E .TR\#8 , "K ! DI6];E 2T SE>* 5G9ES
& DIRECTORIES4 ,DIRECTORIES1 S* Z
,YAH006 ,CANADA 7_+WWW.YAH00.CA_:7 &
A#B;Z 7_+WWW.A2Z.LYCOS.COM_:71 H LI/S (
SITES Y C BR[SE 0SUBJECT4

```

```

,EXAMPLE #B.C.D3 , -PUT] #I
,PROGRAM 7, -PUT] ,BRL ,CODE7

```

```

STRUCT NLIST *NP, *LOOKUP();
CHAR *STRSAVE(), *ALLOC();
INT HASHVAL;

```

```

,EXAMPLE #B.C.E3 ,NONUSE ( TO1
9TO1 & 0,PREC$+ , -PUT] ,BRL ,CODE
,9DICATOR

```

```

,? NUMB] DIVID$ BY _+0_>FF_?16_: IS !
FRAC;N (! SYMBOL TA# US$4

```

```

,NOTE3 ,? EXAMPLE IS 9 HEXADECIMAL
NOT,N4 ,9 ? NOT,N1 ! NUMB]S >E ! DIGITS
#J1 #A1 #B1 #C1 #D1 #E1 #F1 #G1 #H1 #I1
.&! LRS ; ,A1 ; ,B1 ; ,C1 ; ,D1 ; ,E1 ; ,F4
, " !=E1 ! EXAMPLE %[S ! NUMB] _+0FF_: N !
^W _>OFF_:4

```

```

,SEC;N #B.C #AG
,EXAMPLE #B.C.F3 ,NONUSE ( A#I
TO1 9TO1 & 0,PREC$+ , -PUT] ,BRL ,CODE
,9DICATOR 7#E-JJ7

```

```

,,CA/LES,,UNLIMIT$3,= AN 9SI<T 96OLD
CA/LES & PA/ "TS1 H1D TO
_+WWW.CASTLES-OF-BRITAIN.COM_:4

```

```

.#C.J .,,=MAT #AJ-AA
#C.A ,-PUT] ,BRL ,CODE TRANSCRIP;NS
%D USE ! L;GE/ POSSI# BRL L9E ALL[$
0PRODUC;N & DUPLIC,N 3SID],NS4 ,CURR5T
L9E L5G? IS TYPIC,Y #DJ OR #DA CELLS4
#C.B ,A TRANSCRIB]'S NOTE AT ! 2G9N+
( EA* VOLUME (A BOOK %D /ATE ! YE> (
ADOP;N (! ,-PUT] ,BRL ,CODE1 Z WELL Z !
PRIM>Y CODE 9 : ! BOOK IS TRANSCRIB$4
,IF A BOOK REQUIRES M ?AN TWO CODES1 ALL
CODES EMPLOY$ %D 2 LI/$4 7,= O!R 9=M,N
REQUIR$ ON ! ,TRANSCRIB]'S ,NOTES ,PAGE1
SEE AL #D.A1 #AJ.B1 #AC.D1 #AD.A1 &
#AD.B74
#C.C ,SYMBOLS : ! TRANSCRIB] HAS
DEVIS$ OR ^? ASSIGN$ SPECIAL USAGE %D 2
EXPLA9$ OR DRAWN ON A 8,SPECIAL
,SYMBOLS0 PAGE4 ,^! SYMBOLS %D 2
TRANSCRIB$ 9 A3ORD.E )! RULES (! ,BRL
,=MATS ,CODE4 ,NEI /&>D ,-PUT] ,BRL
,CODE SYMBOLS NOR SYMBOLS ( O!R ADOPT$
CODES : >E US$ 9 A3ORD.E )! RULES (!
APPLICA# CODE7S7 %D 2 9CLUD$ ON !
8,SPECIAL ,SYMBOLS0 PAGE1 EXCEPT Z MAY 2
,SEC;N #B.C-C.C #AI
REQUIR$ 0APPLICA# CODES4 A#AA
#C.D ,A BLANK L9E %D PREC$E & FOLL[
PROGRAMS & -PUT] NOT,N SET (F F ! BODY (
! TEXT4 ,TRANSCRIP;N ( S* MAT]IAL %D 2G9
9 CELL #A (! APPROPRIATE L9E1 ) RUNOV]S
2G9N+ 9 CELL #B (! FOLL[+ L9E7S74 7,SEE
,SEC;NS #F.J1 #G.J1 & #I.J = M 9=M,N AB
RUNOV]S47
-----#AB
,EXAMPLE #C.D.A3 ,-PUT] ,NOT,N
,SET ,(F F ,BODY ( ,TEXT

,! FOLL[+ ,,JCL MAY 2 US$ :5 ! ABV
/EPS >E -PLETE1 OR Z A ?IRD /EP (! ABV1
6G5]ATE ! -PIL]3

```

```

//_>LISP _>JOB
//_>STEP3 _>EXEC _>PGM=LISP
//_>STEPLIB _>DD _>DSNAME=LISP,VOLUME_&
=SER_<=ZZZZZZ,_>UNIT_<=WWW,_>DISP=OLD
//_>LISPOUT _>DD _>SYSOUT=A

```

```

#D.D , = #CFJ_/#IA ,US]S ,ONLY
,2C ( H>DW>E DI6];ES ON ! , ,IBM
#CFJ_/#IA1 ! FOLL[+ UPDATE C>DS M/ 2
9CLUD$ DUR+ ! ASSEMBLY ( ' ' ' A#AB
-----#AC
,EXAMPLE #C.D.B3 , -PUT] ,NOT,N
7,WEB ,SITE ,A4RESS7 ,SET ,(F F ,BODY (
,TEXT 7#E-JJ7

,3TACT ,C.TRY ,ACRES ,S5IORS , -MUN;Y
,LTD4 = M 9=M,N4 , ! WEB SITE IS GIV5 2L4

HTTP://WWW.ANGELFIRE.COM/BIZ/CASC/_&
INDEX.HTML

```

```

#C.E , :5"E POSSI#1 AN 5TIRE -PUT]
PROGRAM %D 2 TRANSCRIB$ ON A S+LE BRL
PAGE4 , :5 A PROGRAM IS TOO L;G 6FIT ON A
S+LE PAGE1 ANY ID5TIFY+ LABEL %D 2 C5T]$
ON ! LA/ L9E 7USU,Y L9E #BE7 ( EA* PAGE
6: X APPLIES4 ,A PROGRAM L9E _C 2 DIVID$
2T PAGES4 ,TEXT FOLL[+ A PROGRAM REQUIR+
M ?AN "O BRL PAGE %D 2G9 ON A NEW PAGE1
)! TRANSI;N 6! NEW PAGE REPLAC+ ! BLANK
L9E : ORD9>ILY FOLL[S A PROGRAM4

```

```

,SEC;N #C.C-C.E #BA
,EXAMPLE #C.E.A3 ,PROGRAM #AD-AE
) ,ID5TIFY+ ,LABEL 7#E-JJ7

```

```

,PROGRAM #C-E
8,GUESS ! ,NUMB]0 ,PROGRAM

```

```

% _THE "_GUESS THE _NUMBER" PROGRAM
% _CHOOSES A RANDOM NUMBER _&
BETWEEN _1 AND 99
% AND ALLOWS YOU TO GUESS IT
RANDOMIZE
_*VAR_/ HIDDEN, GUESS: _*INT_/
_*VAR_/ REPLY: _*STRING/(1)
_*PUT_/ "_GUESS THE HIDDEN NUMBER"
_*PUT_/ "_IT IS BETWEEN _&
_1 AND 99 INCLUSIVE"
_*LOOP_/
_*VAR_/ COUNT: _*INT_/ := _0
_*PUT_/ "_DO YOU WANT TO PLAY? _&
ANSWER Y OR N" ..
_*GET_/ REPLY
_*EXIT WHEN_/ REPLY = "N"

```

```

,PROGRAM #C-E
% _CHOOSE A RANDOM NUMBER _& #AF
BETWEEN _1 AND 99
RANDINT (HIDDEN, 1, 99)
_*LOOP_/
_*PUT_/ "_ENTER YOUR GUESS _&
(ANY NUMBER BETWEEN _1 AND 99)" ..
_*GET_/ GUESS
COUNT := COUNT + _1
_*IF_/ GUESS < HIDDEN _*THEN_/
_*PUT_/ "_YOU ARE LOW"
_*ELSEIF_/ GUESS > HIDDEN _*THEN_/
_*PUT_/ "_YOU ARE HIGH"
_*ELSE_/
_*PUT_/ "_YOU GOT IT IN ", _&
COUNT, _" GUESSES"
_*EXIT_/

```

```

_*END IF_/
_*END LOOP_/
_*END LOOP_/

```

```

,PROGRAM #C-E
,SEC;N #C.E #BC
#C.F 7#E-JJ7 ,A BLANK L9E NE$ #AG
N PREC$E OR FOLL[ -PUT] NOT,N )9 TEXT
7"HAF REF]R$ 6Z EMB$D$ -PUT] NOT,N74
,H["E1 EMB$D$ -PUT] NOT,N %D 2
TRANSCRIB$ 9 , -PUT] ,BRL ,CODE4 ,IF
EMB$D$ -PUT] NOT,N W N FIT 5TIRELY ON !
CURRST L9E1 B W FIT ON A :OLE BRL L9E1 ?
NOT,N %D 2G9 ON A NEW L9E4 ,L;G] EMB$D$
NOT,N %D 2G9 ON ! CURRST L9E & 2 DIVID$
AT A LOGICAL PLACE S* Z FOLL[+ PUNCTU,N
OR A -PLETE ^W4 ,RUNOV]S %D 2 WRITT5 AT
! M>G9 CURRSTLY 9 E6ECT4 ,TEXT : FOLL[S
EMB$D$ -PUT] NOT,N MAY RESUME ON !
CURRST L9E4 ,! 2G9 , -PUT] ,BRL ,CODE
9DICATOR M/ 2G9 EMB$D$ -PUT] NOT,N
EXCEPT :5 ! F/ SYMBOL ( S* NOT,N IS !
CAPS LOCK 9DICATOR1 ! 2G9 EMPHASIS
9DICATOR1 ! 2G9 ,NEME? ,CODE 9DICATOR1
OR ! 2G9 %APE 9DICATOR4 ,A T]M9,N
9DICATOR M/ ALW 2 US$ )9 EMB$D$ -PUT]
NOT,N 6SIGNAL A RETURN 6PRIM>Y CODE
&_/OR 6C.EL ! E6ECT (! MO/ REC5T 9/.E 9
: ! T]M9,N 9DICATOR APPLIES4 7,= 4CUS.NS
(! CAPS LOCK1 EMPHASIS1 ,NEME? ,CODE1
%APE1 & T]M9,N 9DICATORS1 SEE ,SEC;NS
#D.J1 #AJ.J1 #AB.J1 #AC.J1 & #AG
#AF.J1 RESPECTIVELY47
,EXAMPLE #C.F.A3 ,EMB$D$ , -PUT]
,NOT,N

```

```

,6SPECIFY ALL FILES "ND _>ACCOUNTS_:1
REG>D.S ( EXT5.N1 TYPE ! FOLL[+3

```

```

,EXAMPLE #C.F.B3 ,EMB$D$ , -PUT]
,NOT,N ) ,RUNOV]

```

```

,! MACRO -M& W *ANGE ALL O3URR;ES (
_>ROM TO _>RAM_:4 ,IF US$ ON ! EXAMPLE 9
! ;,J -M&S1 ! RESULT+ L9E W R1D3
_>WHEN _>IN _>RAM _>DO _>AS _>THE _&
_>RAMANS _>DO, _>AND _>BE _&
_>RAMANTIC..._:
-----#AH

```

```

,NOTE3 ,! ^WS ,,ROM & ,,RAM >E "P (A
4CUS.N ( TEXT $IT+ & >E N US$ Z
ACRONYMS4

```

```

,SEC;N #C.F #BE
,EXAMPLE #C.F.C3 ,EMB$D$ A#AH
, -PUT] ,NOT,N 7,WEB ,SITE7 ) ,RUNOV]
7#E-JJ7

```

```

,6GET M 9=M,N AB ,,IBM'S ,,PC OP;NS1 =
,,IBM & O!R ,,PC'S1 VISIT \R WEB SITE AT
_+WWW.CAN.IBM.COM/PC/CA/SOLUTIONS/_&
HOME.HTML_:1 OR YR LOCAL RETAIL]4

```

```

,EXAMPLE #C.F.D3 ,EMB$D$ , -PUT]

```

,NOT,N ON ,NEW ,L9E 7#E-JJ7

,,COBOL
 ,-MON ,BUSI;S ,ORI5T\$,LANGUAGE GIVES
 9/RUC;NS 6! -PUT] 0US+ ,5GLI%-L S5T;ES
 S* Z3
 _>ADD _>SALES-AMT _>TO _>TOTAL-AREA_:4
 ,,COBOL PROGRAMS >E EASY 6R1D1 B REQUIRE
 A LOT (SPACE4

..#D.J ,,UPP] ,,& ,,L[] ,,CASE #AI
 .,,DESIGN,N
 #D.A ,UN.S O!RWISE 9DICAT\$1 ! , -PUT]
 ,BRL ,CODE ASSUMES T LRS & ^WS 3TA9\$ 9
 -PUT] NOT,N >E L[] CASE4 ,:5 -PUT] NOT,N
 IS PREDOM9ANTLY OR EXCLUSIVELY UPP] CASE
 7E4G41 A ,,BASIC TEXT71 ! TRANSCRIB] MAY
 ELECT 6USE S+LE-CELL LRS 69DICATE UPP]
 CASE & LRS PREC\$\$ 0DOTS #DEF1 ! %IFT
 9DICATOR1 6DESIGNATE L[] CASE4 ,IF -PUT]
 NOT,N "?\T A TEXT IS EXCLUSIVELY UPP]
 CASE1 A TRANSCRIB]'S NOTE 6? E6ECT %D
 APPE> ON ! 8,TRANSCRIB]'S ,NOTES0 PAGE4
 7,EXAMPLES #A.A.B1 #A.B.A1 #A.C.A;D1
 #A.D.A1 #A.G.A1 #E.A.A1 #F.C.A1 #H.A.A1
 #I.A.A1 & #AD.B.A WD 9CLUDE S* A NOTE47
 ,IF -PUT] NOT,N "?\T A TEXT IS N
 EXCLUSIVELY UPP] CASE1 ! TRANSCRIB]'S
 NOTE %D IMMLY PREC\$E ! MAT]IAL 6: X
 APPLIES4

,SEC;N #C.F-D.A #BG
 ,EXAMPLE #D.A.A3 ,S+LE A#AI
 ,,BASIC ,PROGRAM ,TRANSCRIB\$)\T ,CASE
 ,9DICATORS
 ,',! FOLL[+ ,,BASIC PROGRAM IS
 5TIRELY 9 UPP] CASE & W 2 TRANSCRIB\$
)\T CASE 9DICATORS4,'

10 INPUT "YOUR FIRST NAME";A\$
 20 INPUT "YOUR MIDDLE NAME";B\$
 30 INPUT "YOUR LAST NAME";C\$

#D.B ,%IFT ,9DICATOR _ 7#DEF73 ,!
 %IFT 9DICATOR PREC\$ES A S+LE UPP] CASE
 LR APPE>+ 9 EMB\$D\$ -PUT] NOT,N &_/OR
 -PUT] NOT,N SET (F F ! BODY (! TEXT : IS
 O!RWISE 9 L[] CASE4 ,TWO OR M UPP] CASE
 LRS 03URR+ 9 PREDOM9ANTLY L[] CASE TEXT
 --:E!R ^! LRS APPE> 3SECUTIVELY OR >E
 SEP>AT\$ 0NUMB]S1 PUNCTU,N1 ETC4--M/ 2
 PREC\$\$ 0! CAPS LOCK 9DICATOR 7SEE ,SEC;N
 #D.C474

-----#BJ
 #D.C ,CAPS ,LOCK ,9DICATOR _>
 7#DEF1 #CDE73 ,! CAPS LOCK 9DICATOR IS
 US\$ 9 9/.ES 9 : EMB\$D\$ -PUT] NOT,N &_/OR
 -PUT] NOT,N SET (F F ! BODY (! A#BJ
 TEXT APPE>S 9 A -B9,N (UPP] & L[] CASE4
 ,! CAPS LOCK 9DICATOR REMA9S 9 E6ECT
 UNTIL C.ELL\$ 0! CAPS REL1SE 9DICATOR OR


```
! NEXT SIGNIFICANT SPACE1 :I*"E -ES F/4
,NEI CELLS 03UPI$ 0! 3T9U,N 9DICATOR1
SPACES FOLL[+ ! 3T9U,N 9DICATOR1 NOR !
SPACE %[+ 9D5T,N (A RUNOV] >E 3SID]$
SIGNIFICANT SPACES4
,EXAMPLE #D.C.A3 ,3SECUTIVE &
,S+LE ,UPP] ,CASE ,LRS
```

```
_>UNDEFINED; $(_IMPLEMENTATION DETAIL)
```

```
,EXAMPLE #D.C.B3 ,CASE ,DESIGN,N 9
;,E-MAIL ,A4RESSES 7#E-JJ7
```

```
,USU,Y LRS 9 ;E-MAIL A4RESSES >E ALL 9
L[] CASE2 H["E1 AN A4RESS ON "O LI/ IS
+_RED2@NETCOM.CA_:4
```

```
,SEC;N #D.A-D.C #BI
#D.D ,CAPS ,REL1SE ,9DICATOR B#BJ
_< 7#DEF1 #ABF73 ,! CAPS REL1SE 9DICATOR
C.ELS ! E6ECT (! CAPS LOCK 9DICATOR4
,EXAMPLE #D.D.A3 ,CAPS ,LOCK &
,CAPS ,REL1SE ,9DICATORS
```

```
_>VFUN _>PST_<MSG_TO_VEC(IPC_MSG
```

```
..#E.J ,,REPRESENT,N ,( ,,3TROL #BA
.,,"*S
#E.A ,3TROL "*S1 :E!R 9 ,,ASCII OR
O!R -PUT] CODES1 >E N ACTU,Y PR9T$
0-PUT]S & >E "!=E SYMBOLIZ$ 9 PR9T 9 A
V>IETY ( WAYS4 ,= EXAMPLE1 ! C>RIAGE
RETURN FUNC;N MAY 2 REPRESENT$ BY
8,,RETURN10 8,,5T]10 8,,CR10 A D[NW>D
LEFT-CURV$ >R[1 A DEPICT$ KEY1 OR 9
V>I\ S O!R WAYS4 ,?US1 ! BRL TRANSCRIP;N
%D FOLL[ ! PR9T SYBOLOGY LIT],Y1 9
A3ORD.E )! O!R RULES ( ? CODE4
,EXAMPLE #E.A.A3 ,C>RIAGE ,RETURN
,REPRESENT,N
```

```
) CREATE/I DSWITCH.CLI_$CR_:
```

```
,NOTE3 ,! D[NW>D LEFT-CURV$ >R[ IS US$ Z
A C>RIAGE RETURN4 ,"!E1 9 ? EXAMPLE !
LRS CR >E US$ )9 ! %APE 9DICATORS4
,EXAMPLE #E.A.B3 8,C>RIAGE
,RETURN0 ,REPRESENT,N
```

\$K>RETURN_:
 ,SEC;N #D.D-E.A #CA

.#F.J .,,3T9U,N .,,9DICATOR #BB-BC
 #F.A ,3T9U,N ,9DICATOR _&
 7#DEF1 #ABCDF73 ,! 3T9U,N 9DICATOR IS A
 L9E DIVI.N SIGN4 ,X IS US\$ 9 9/.ES 9 : !
 L9E> >RANGE;T (! PR9T IS SIGNIFICANT &!
 BRL L9E IS N L;G 5 6F9I% ! PR9T L9E4
 ,L9E DIVI.N MAY O3UR)9 EMB\$D\$ -PUT]
 NOT,N Z WELL Z)9 -PUT] NOT,N SET (F F !
 BODY (! TEXT & MAY G5],Y O3UR AT 3V5I5T
 PO9TS 7I4E41 ALL AVAILA# SPACES ON ! BRL
 L9E NE\$ N 2 FILL\$74 ,IF NO SPACES >E
 PRES5T AT ! PO9T (L9E DIVI.N1 ! 3T9U,N
 9DICATOR %D IMMLY FOLL[! LA/ CELL (!
 BRL L9E 7I4E41 NO SPACES %D 9T]V5E74 ,!
 3T9U,N 9DICATOR MAY N 2 US\$ 2T ! CELLS (
 A MULTI-CELL SIGN 7E4G41 2T ! F/ &
 SECOND CELLS (! 8LEFT BRACE0 SYMBOL74
 ,:5 US\$ 6DIVIDE C.TA# SPACES 9DICATORS1
 ! 3T9U,N 9DICATOR M/ 2 PREC\$\$ 0TWO FULL
 CELLS 7#ABCFE1 #ABCDEF74 ,! 3T9U,N
 9DICATOR IS N 3SID]\$ A C.TA# SPACE4
 7,SEE ,SEC;N #I.J =A 4CUS.N (! C.TA#
 SPACES 9DICATOR & XS USE47

,SEC;N #F.J-F.A #CC
 ,EXAMPLE #F.A.A3 ,3T9U,N A#BC
 ,9DICATOR

_>OFUN _>PST_<IPCQ_PUT(SEID SENDER,_&
 RECEIVER;EV_TYPE EV;_>BOOLEAN _&
 PSEUDO_INT;

#F.B ,! REMA9D] (! ORIG9AL PR9T L9E
 IS 3T9U\$ ON ! NEXT BRL L9E7S74 ,RUNOV]S
 (-PUT] NOT,N SET (F F ! BODY (! TEXT
 2G9 9 CELL #B4 ,RUNOV]S (EMB\$D\$ -PUT]
 NOT,N 2G9 AT ! M>G9 CURR5TLY 9 E6ECT 9
 ORD] 6AID ! SCANN+ BRL R1D] 9 4T+UI%+
 PROGRAM F N>RATIVE4

-----#BD
 #F.C ,A 3T9U,N 9DICATOR %D 2 US\$:5
 ! 5TIRE PR9T L9E W N FIT ON ! BRL

L9E7S71 EV5 IF ! L9E DIVI.N O3URS FOLL[+
A -PLETE ^W4

,EXAMPLE #F.C.A3 ,3T9U,N A#BD
,9DICATOR)9 ,EMB\$D\$, -PUT] ,NOT,N

#AB4 , -PLETE ! FOLL[+ PROGRAM1 : IS
SIMIL> 6T (FRAME #AA4 ,USE /R+ V>IA#S
= SUB/R+S 6PRODUCE ? 4PLAY F _+\$Z = _&
"THE END OF A STRING IS ON THE _&
RIGHT." _:

,EXAMPLE #F.C.B3 ,3T9U,N ,9DICATOR
)9 ,EMB\$D\$,WEB ,SITE ,A4RESS 7#E-JJ7

,MO/ (^! PAGES H A L9K 6! .,SATUR"D
.,NI<T .,LIVE SKET* ,CNN REL5T.SLY AIR\$
(,H>TMAN IMP]SONAT+ ,PRESID5T ,CL9TON2
Y C D[NLOAD X DIRECTLY F _+WWW.CNN._&
COM/_>SHOWBIZ/TV_</9805/28/HARTMAN._&
OBIT/PHIL.HARTMAN.43.3.8.MOV_:4

,SEC;N #F.A-F.C #CE

.#G.J .,,9D5T\$.,,MAT]IAL #BE
#G.A ,:5 SU3ESSIVE LEVELS (9D5T,N
O3UR 9 PR9T1 BRL L9ES %D AL 2 9D5T\$4
,EA* BRL LEVEL (9D5T,N %D 2G9 TWO
SPACES 6! "R (! PREVI\S LEVEL1 REG>D.S (!
PR9T SPAC+ US\$4 ,?US1 SU3ESSIVE LEVELS
(BRL 9D5T,N 2G9 9 CELLS #A1 #C1 #E1
ETC4 ,:5 9 D\BT1 3SID] SPACES AT ! LEFT
Z LEVELS (9D5T,N1 N C.TA# SPACES4
#G.B ,RUNOV]S %D ALW 2G9 9 CELL #B1
REG>D.S (! CURR5T DEP? (9D5T,N4

```

,EXAMPLE #G.B.A3 ,9D5T$ ,MAT]IAL

RESULTS() /* PRINT TABLE USING AR_&
RAY POINTERS */
-[
  EXTERN INT *P_A1,*P_A2,*P_A3;
  INT I;
  PRINTF("\N%10S%10S%10S%10S\N\N",_&
"I", "_A1[I]", "_A2[I]", "_A3[I]");
  FOR(I = 0; I <= _>MAX-1; _<I++)
    PRINTF("%10D%10D%10D%10D\N",I,_&
P_A1[I],P_A2[I],P_A3[I]);
-]
,SEC;N #G.J-G.B #CG
,EXAMPLE #G.B.B3 ,9D5T$ #BF
,MAT]IAL

```

```

_UNION_ORDER_PACKAGE =
-[ &_COMPARE_>AB _$0_:
  (LAMBDA (X Y)
    (COND ((&_MEMBER_A X) (COND _&
((&_MEMBER_A Y) (&_COMPARE_A X Y))
(T -1)))
    ((&_MEMBER_A Y) 1)
    (T (&_COMPARE_B X Y))))_)

```

```

,NOTE3 ,! SYMBOL =! "R-P09T+ >R[ 0
SELECT$ F ! ,NEME? ,CODE & 5CLOS$ )9 !
%APE 9DICATORS4

```

```

..#H.J , ,ISOLAT$ , ,L[]-CELL #BG
.,,SIGNS
#H.A ,:5 A L[]-CELL SIGN IS ISOLAT$
7I4E41 PREC$$ & FOLL[$ 0A SPACE OR L9E
5D71 DOTS #DEF M/ IMMLY PREC$E ? SIGN4
,! L[]-CELL SIGN MAY 2 A S+LE-DIGIT
NUMB]1 QUOT,N M>K1 APO/ROPHE1 -MA1
HYPH5_/M9US SIGN1 OR SEMICOLON4 ,!
RESULT+ TWO-CELL SYMBOL SIGNIFIES A S+LE
PR9T ""4
,EXAMPLE #H.A.A3 ,ISOLAT$ ,L[]-
,CELL ,SIGNS

```

```

107 : CLEAR YLEN _0 DO XLEN _0 DO _&
_0 I J UNIVERSE C! LOOP LOOP _;

```

```

,SEC;N #G.B-H.A #CI

```

```

_ ; +-+--+--+--+--+--+--+--+--+--+--+--+_&
  -+--+--+--+--+--+_===== :
_ ; _\ _===== _>INDEX _===== _&
  ===== +_>TI+ _>RPL + _===== :
_ ; +-+--+--+--+--+--+--+--+--+--+--+--+_&
  -+--+--+--+--+--+_===== :

```

```
,EXAMPLE #I.B.B3 ,C.TA#          #CA
,SPACES ,2G9N+ ON ,RUNOV] ,L9E
```

```
_6 : <_A>  _0 _0 _>HERE 1+ _>CON_&
VERT _>DROP _>DROP ( EXTRACT VARIA_&
BLE # )
_7 _===== [_>COMPILE] _>LITERAL_&
_===== ( MAKE IT A LITERAL )
_8 _===== _>COMPILE <_>ADDR> ;_&
_===== ( DYNAMIC PART OF ADDRESS )
_9 _1 _>WIDTH ! ( SET SIGNIFICANT _&
CHARACTERS IN NAME )
10 : @_X _===== ( --- VALUE ) _&
( FETCH CONTENTS OF VARIABLE _X )
```

```
,SEC;N #I.B          #DC
```

```
.#AJ.J .,,EMPHASIS          #CB-CC
.,,9DICATORS
  #AJ.A ,2G9 ,EMPHASIS ,9DICATOR _*
7#DEF1 #AF72 ,5D ,EMPHASIS ,9DICATOR _/
7#DEF1 #CD73 ,EMPHASIS 9DICATORS 5CLOSE
-PUT]-RELAT$ MAT]IAL : IS EMPHASIZ$ =
TE*NICAL R1SONS 7E4G41 64T+UI% :AT A
/UD5T TYPES F :AT ! -PUT] RESPONDS74
,PR9T ME?ODS ( EMPHASIZ+ ? MAT]IAL MAY
9CLUDE "UL9+1 ITALICIZ+1 *ANG+ TYPE
FONTS1 *ANG+ COLOR1 ETC4
  ,EXAMPLE #AJ.A.A3 ,EMPHASIS
,9DICATORS
```

```
''' = EXAMPLE1 _*CHKDSK A:*. * /V_/_: WD
*ECK ! DISKETTE 9 DRIVE ;,A1 *ECK ALL
FILES ON X = FRAG;T,N1 & 4PLAY ! "NS (
ALL FILES & DIRECTORIES4
```



```
FILL >E N REPRES5T$ 9 ! , -PUT] ,BRL
, CODE1 PR9T COLUMN =M IS N MA9TA9$4 , " :
ITEMS >E PRES5T 9 COLUMN #B !Y >E
SEP>AT$ F ITEMS 9 COLUMN #A 0#C SPACES4
```

```
..#AA.J , , -PUT] , , BRL , , CODE #CG
, , 9DICATORS
#AA.A , 2G9 , -PUT] , BRL , CODE
, 9Dicator _+ 7#DEF1 #CDF72 , 5D , -PUT]
, BRL , CODE , 9Dicator _: 7#DEF1 #AEF73
, -PUT] , BRL , CODE 9DICATORS NE$ N 5CLOSE
-PUT] NOT,N SET (F F ! BODY (! TEXT4
, -PUT] , BRL , CODE %D 2 5T]$ :5 TEXTUAL
PASSAGES 9CLUDE OBVI\S -PUT] NOT,N
7E4G41 MAT]IAL 6BE TYP$ 6! -PUT] 0! R1D]
9 A SPECIFIC MANN]74
```

```
#AA.B , IF ! CAPS LOCK 9Dicator1 !
2G9 EMPHASIS 9Dicator1 ! 2G9 , NEME?
, CODE 9Dicator1 OR ! 2G9 %APE 9Dicator
2G9S EMB$D$ , -PUT] , BRL , CODE1 ! 2G9
, -PUT] , BRL , CODE 9Dicator NE$ N 2 US$2
H["E1 ! 5D , -PUT] , BRL , CODE 9Dicator M/
ALW 2 US$ 6SIGNAL A RETURN 6REGUL> TEXT4
, ! PURPOSE ( , -PUT] , BRL , CODE 9DICATORS
IS 6CL>IFY POT5TI,Y AMBIGU\S SITU,NS
7E4G41 ! "*" PREC$+ ;S 9 ! FILE"N
_+ITEM.S_: CD 2 EI A P]IOD OR "P (!
8LESS0 SIGN74 , FREQU5T SWIT*ES 2T , -PUT]
, BRL , CODE &! PRIM>Y CODE 2+ US$ %D 2
, SEC;N #AJ.B-AA.B #DI
AVOID$1 "PICUL>LY )9 ! SAME BRL A#CG
L9E4 , ?US1 UN.S A R[ OR COLUMN (A TA#1 A
TITLE1 OR O!R 3TIGU\S MAT]IAL C 2
TRANSCRIB$ 5TIRELY 9 ! PRIM>Y CODE1
, -PUT] , BRL , CODE %D 2 EMPLOY$4
, EXAMPLE #AA.B.A3 , -PUT] , BRL
, CODE , 9DICATORS
```

```
''' ! SYMBOLS (! =M _+*1, *2,_: ETC4
```

```
, EXAMPLE #AA.B.B3 , -PUT] , BRL
, CODE , 9DICATORS 7#E-JJ7
```

```
, AN ;E-MAIL A4RESS : USES UPP] CASE LRS
IS _>ROGERJ_<@EAST.CNIB.CA_:4
```

```
-----#CH
, EXAMPLE #AA.B.C3 , AVOID+ , FREQU5T
, SWIT*ES 2T , -PUT] , BRL , CODE & , PRIM>Y
, CODE
```

```
''' ! SYMBOLS (! =M _+*1 AND *2_:1 ETC4
```

```
, EXAMPLE #AA.B.D3 , USE ( A#CH
, -PUT] , BRL , CODE , 9DICATORS 6,CL>IFY
, AMBIGUITIES
```

```
_+ITEM.S_:
```


,NOTE3 ,? EXAMPLE APPE>\$)9 ! BODY (!
 TEXT4 ,_H ! -PUT] NOT,N BE5 SET (F F !
 BODY (! TEXT1 , -PUT] ,BRL ,CODE
 9DICATORS WD H BE5 UNNEC4

,SEC;N #AA.B #EA

..#AB.J ,,NEME? ,,CODE #CI
 .,,9DICATORS
 #AB.A ,2G9 ,NEME? ,CODE ,9DICATOR _%
 7#DEF1 #ADF72 ,5D ,NEME? ,CODE ,9DICATOR
 _: 7#DEF1 #AEF73 ,NEME? ,CODE MAY 2 NEC
 :5 PURELY MA!MATICAL EXPRES.NS 7E4G41
 =MULAS1 EQU,NS1 ! MULTIPLIC,N SYMBOL :5
 REPRESENT\$ 0A DOT1 ETC47 >E 6BE
 TRANSCRIB\$ OR :5 NO APPROPRIATE , -PUT]
 ,BRL ,CODE SYMBOL EXI/S 7E4G41 " ! IS NO
 HORIZONTAL FRAC;N 9DICATOR 9 , -PUT] ,BRL
 ,CODE74 ,H["E1 ,NEME? ,CODE %D 2 RESORT\$
 6ONLY R>ELY1 &A ,NEME? ,CODE EXP]T %D 2
 3SULT\$ 2F ! TRANSCRIP;N IS "UTAK54 ,IF
 FREQU5T USE (,NEME? ,CODE SEEMS NEC1 !
 PRIM>Y CODE 9 : ! TEXT IS TRANSCRIB\$ MAY
 NE\$ 6BE *ANG\$4

,SEC;N #AB.J-AB.A #EC

```

,EXAMPLE #AB.A.A3 ,NEME? #DJ
,CODE ,9DICATORS
, ',PR9T COLUMN =M *ANG$ Z FOLL[S3
,MA!MATICAL ,NOT,N
,,BASIC ,NOT,N, '

```

```

_%A1+A2+A3+A4_:
_>A(1)+A(2)+A(3)+A(4)
_%X1" ^2"+X2" ^2_:
_>X(1)^2+X(2)^2
_%(A-R1)(Z-R2)_:
_(>A-R(1))*(Z-R(2))
_%1.5(Z;I+1"-Z;I")_:
1.5*(>Z(I+1)-Z(I))
-----#DA

```

```

,EXAMPLE #AB.A.B3 ,NEME? ,CODE
,9DICATORS ) ,HORIZONTAL ,FRAC;N ,L9E

```

```

''' SITU,N IS CL>IFI$ IF "O 3SID]S A
DI6]5T =M (! EQU,N (! ELLIPSE3

```

```

_%?X^2"/A^2"#+?Y^2"/B^2"# .K #1_:

```

```

": ;A & ;B >E ! RADII AL;G ! ;X & ;Y '''

```

```

#AB.B ,NEME? ,CODE 9DICATORS A#DA
%D 2 WRITT5 UNSPAC$ F ! MAT]IAL !Y
5CLOSE4 ,)9 ,NEME? ,CODE 9DICATORS1
,NEME? ,CODE NOT,N & RULES %D 2 FOLL[$4
,IF A L;G PASSAGE 9CLUDES S"EAL ,NEME?
,CODE EXPRES.NS1 ! 2G9 ,NEME? ,CODE
9DICATOR %D PREC$E ! F/ ,NEME? EXPRES.N4
,! 5D ,NEME? ,CODE 9DICATOR %D 2 WRITT5
AF ! LA/ ,NEME? EXPRES.N 9 ! PASSAGE HAS
BE5 TRANSCRIB$4 ,H["E1 , -PUT] ,BRL ,CODE
9DICATORS %D 5CLOSE -PUT] NOT,N )9
,NEME? ,CODE EXPRES.NS4 ,! 5D ,NEME?
,CODE 9DICATOR RETURNS ! TRANSCRIP;N 6!
BASEL9E ( WRIT+4

```

```

,EXAMPLE #AB.B.A3 ,NEME? ,CODE
,9DICATORS )9 , -PUT] ,BRL ,CODE

```

```

,N[ WE W CL"ELY 9ITIALIZE \R A4$ V>IA#
_+D__NORM TO _%2.B 2F 5T]+ ! LOOP1 S E
PASS WE ADD #4.B_: TO D__NORM_: 2F A4+
X1 9 TURN1 6! =M] VALUE ( ;D4

```

```

,SEC;N #AB.A-AB.B #EE

```

```

.#AC.J .,,%APE .,,9DICATORS      #DB-DC
.#AC.A ,2G9 ,%APE ,9DICATOR _$
7#DEF1 #ABDF72 ,5D ,%APE ,9DICATOR _:
7#DEF1 #AEF73 ,%APE 9DICATORS >E US$
65CLOSE A BRIEF S]IES ( SYMBOLS DESCRIB+
A %APE N REPRES5T$ 9 ! , -PUT] ,BRL
,CODE4
,EXAMPLE #AC.A.A3 ,%APE ,9DICATORS

_S _$S-_: _R =_$LB_: < S, ADD > _\ _&
S _S _- _R_$RB_:

,NOTE3 ,! %APE =! SQU>E ) AN 9T]IOR M9US
0 TRANSCRIB$ Z AN ;S FOLL[$ 0A M9US &
5CLOS$ )9 ! %APE 9DICATORS4 ,! LEFT & "R
D\#-B>R$ BRACKETS 7 TRANSCRIB$ Z LB & RB
)9 ! %APE 9DICATORS4
,EXAMPLE #AC.A.B3 ,%APE ,9DICATORS
7#E-JJ7
, ' ,IN ! MAT]IAL 2L A CLOCK ICON IS
%[N 0! LRS CL TRANSCRIB$ )9 %APE
9DICATORS4, '

,6A3ESS ! S*$UL]1 CLICK ON ! _$CL_:_:
,SEC;N #AC.J-AC.A      #EG
ICON 9 ! ,ANTI-,VIRUS ,TOOLKIT      A#DC
W9D[4
.#AC.B ,:5 KEYS ON A -PUT] KEYBO>D >E
DEPICT$1 ! 2G9 %APE 9DICATOR %D 2 FOLL[$
0! LR ;K1 ! KEY LEG5D1 &! 5D %APE
9DICATOR4
-----#DD
,EXAMPLE #AC.B.A3 ,DEPICT$ ,KEY

,HIT _$K_>ENTER_:_: KEY4

.#AC.C ,TRANSCRIB]S FAMILI> ) ,NEME?
,CODE MAY SELECT OR DEVISE SYMBOLS 9
A3ORD.E ) ,NEME? ,CODE 3V5;NS1 ?US
5HANC+ 3SI/5CY AM;G BRL CODES4
,EXAMPLE #AC.C.A3 ,USE ( ,NEME?
,CODE ,SYMBOL ( ,%APE )9 ,%APE
,9DICATORS

,"*S EXI/+ ONLY 9 ! ;,A SET3

& # % _$4_: @

,EQUIVAL5T "*S F '''

,NOTE3 ,! SYMBOL =! SQU>E 0      A#DD
SELECT$ F ! ,NEME? ,CODE & 5CLOS$ )9 !
%APE 9DICATORS4
.#AC.D ,A TRANSCRIB]'S NOTE AT !
2G9N+ (! "W OR IMMLY PRIOR 6! F/ USE (
%APE 9DICATORS %D EXPLA9 ! SYMBOL'S M1N+
7IF "KN71 OR XS %APE %D 2 DESCRIB$ OR
DRAWN4 7,SEE ,EXAMPLE #AC.A.B47

```

,SEC;N #AC.A-AC.D #EI

```

..#AD.J ,,TRANSCRIB]'S ,,OP;N #DE
,,SYMBOLS
#AD.A ,PRIM>Y ,TRANSCRIB]'S ,OP;N
,SYMBOL _! 7#DEF1 #BCDF72 ,SECOND>Y
,TRANSCRIB]'S ,OP;N ,SYMBOL _
7#DEF1 #DF73 ,:5 A "*" N REPRES5T$ 0!
,-PUT] ,BRL ,CODE 03URS 9 A TRANSCRIP;N
) GRT FREQU5CY1 ! TRANSCRIB] MAY USE A
TRANSCRIB]'S OP;N SYMBOL 6REPRES5T T "*"4
,! M9IMUM NUMB] ( SPACES REQUIR$
6PRODUCE ! "*" IS ?US REDUC$ F FIVE 6TW04
,TRANSCRIB]'S OP;N SYMBOLS %D 2 EXPLA9$
9 A TRANSCRIB]'S NOTE4
,EXAMPLE #AD.A.A3 ,TRANSCRIB]'S
,OP;N ,SYMBOL ,REPRES5T+ ,NON-,ASCII
,"*
',IN ! EXAMPLE : FOLL[S1 ! NON-
,,ASCII "*" : IS REPRES5T$ 0! PRIM>Y
TRANSCRIB]'S OP;N SYMBOL IS ! LR ;B
)A SLA% DRAWN "? X4,'

```

```

_>RENAME_<_! *FILENAME1_/[_*/EXT1_/]&
[_*.PSW_/][_*:D_/]_!>TO_<_! *FILE_&
NAME2_/[_*/EXT2_/]
,SEC;N #AD.J-AD.A #FA
#AD.B ,:5 ! PR9T TEXT DOES N A#DE
9CLUDE A "*" US$ 0! ,,-PUT] ,BRL ,CODE B
DOES 9CLUDE A "*" N REPRES5T$ 0! ,,-PUT]
,BRL ,CODE1 ! M1N+ ( AN UNUS$ ,,-PUT]
,BRL ,CODE "*" MAY 2 *ANG$ DUR+ T
TRANSCRIP;N 7E4G41 9 AN ,,EBCDIC
TRANSCRIP;N1 ! UNUS$ ,,-PUT] ,BRL ,CODE

```

TILDE CD REPRES5T ! FREQU5TLY US\$
 ,,EBCDIC 8N0 SIGN74 ,A TRANSCRIB]'S NOTE
 %D EXPLA9 S* SUB/ITU;NS4
 ,EXAMPLE #AD.B.A ,TRANSCRIB]'S
 ,SUB/ITU;N 7,,ASCII ,TILDE ,2COMES
 ,,EBCDIC 8,N0 ,SIGN7

EX = (P _^= NULL);

.#AE.J .,,HALF-L9E .,,%IFTS #DF-DG
 #AE.A ,HALF-,L9E ,%IFT ,UP _#
 7#DEF1 #CDEF72 ,HALF-,L9E ,%IFT ,D[N _?
 7#DEF1 #ADEF72 ,5D ,HALF-,L9E ,%IFT _:
 7#DEF1 #AEF73 ,HALF-L9E %IFTS UP
 REPRES5T SUP]SCRIPTS--& 9 "S CASES
 EXPON5TI,N--: TYPIC,Y APPE>)9 TEXT4 ,IF
 EXPON5TI,N)9 TEXT IS 9DICAT\$ 0! UP >R[1
 ! UP >R[SYMBOL 7DOTS #DE ^7 %D 2
 TRANSCRIB\$4 ,HALF-L9E %IFTS UP T]M9ATE
 EI)A T]M9,N 9DICATOR 7SEE ,SEC;N #AF.J7
 OR AT ! NEXT SPACE4 ,:5 ! SUP]SCRIPT IS
 N FOLL[\$ IMMLY 0A SPACE1 ! T]M9,N
 9DICATOR M/ 2 US\$4

#AE.B ,HALF-L9E %IFTS D[N >E US\$ =
 TRANSCRIB+ SUBSCRIPTS & EXPRES.NS 9 : !
 NUMB] BASE APPE>S 2L NORMAL L9E LEVEL4
 ,HALF-L9E %IFTS D[N T]M9ATE EI)A T]M9,N
 9DICATOR 7SEE ,SEC;N #AF.J7 OR AT ! NEXT
 SPACE4 ,:5 ! SUBSCRIPT IS N FOLL[\$ IMMLY
 0A SPACE1 ! T]M9,N 9DICATOR M/ 2 US\$4

,SEC;N #AD.B-AE.B #FC
 ,EXAMPLE #AE.B.A3 ,HALF-,L9E #DH
 ,%IFTS

#G.B.B .,DATA .,9ITIALIZ,N .,/ATE;T4 ,A
 DATA 9ITIALIZ,N /ATE;T IS (! =M3

>DATA K?1_:/D_?1_:/,K_?2_:/_&
 D_?2_:/,...,K_?N_:/D_?N_:/

":3

7#A7 ,EA* ;K IS A LI/ 3TA9+ "NS (V>IA#S
 & >RAY ELE;TS4
 7#B7 ,EA* ;D IS A LI/ (3/ANTS & OP;N,Y
 SIGN\$ 3/ANTS4

,EXAMPLE #AE.B.B3 ,HALF-,L9E ,%IFT
 ,T]M9AT\$ 0A ,SPACE

''' ,A LI/5] MAY SET ,NDAC L[7%[N 2T
 +T?5 AND T_?6_:_:7 B1 X M/ D S 2F
 ,,NRFD IS SET HI< AT _+T_?6_:_:4

#AE.C ,REF];E 9DICATORS #DI

DESIGNAT+ FOOTNOTES >E TRANSCRIB\$ 9 !
 PRIM>Y CODE 2+ US\$1 N Z HALF-L9E %IFTS
 UP4

,EXAMPLE #AE.C.A3 ,REF];E
 ,9DICATOR 7,5GLI% ,BRL7

,? *APT] 4CUSSES ! USE (! ,PASCAL 99
 99 ,AT L1/ #AFJ;,K BYTES (MA9
 /ORAGE >E REQUIR\$ 69/ALL & USE
 ,,DOS ,PASCAL4
 SUBSY/EM T OP]ATES "U ! 5VIRON;T '''

,EXAMPLE #AE.C.B3 ,REF];E
 ,9DICATOR 7,NEME? ,CODE7

10 _>PRINT
 20 _>PRINT3
 99 _>END

>RUN% @]#4
 @]#4 ,IF Y >E N US+ AN 9T]ACTIVE
 SY/EM1 IGNORE ! ^W ,,RUN %[N 2F
 EA* \TPUT4_:
 ,SEC;N #AE.B-AE.C #FE

.#AF.J .,,T]M9,N .,,9DICATOR #EJ-EA
 #AF.A ,T]M9,N ,9DICATOR _:
 7#DEF1 #AEF73 ,! T]M9,N 9DICATOR IS
 3TEXT DEP5D5T & T]M9ATES EMB\$D\$ -PUT]
 NOT,N1 ,NEME? ,CODE1 A %APE 9DICATOR1
 HALF-L9E %IFTS UP1 & HALF-L9E %IFTS D[N4
 ,IF T]M9,N 9DICATORS M/ 2 NE/\$ 7I4E41 IF
 "O OR M T]M9,N 9DICATORS 9T]V5E 2F !
 E6ECT (! F/ LAPSES71 ! F/ T]M9,N
 9DICATOR C.ELS ! E6ECT (! MO/ REC5T
 SYMBOL 6: X APPLIES4 ,= EXAMPLE1 EMB\$D\$
 -PUT] NOT,N 9CLUD+ AN 5T]+ -PUT] CODE
 9DICATOR 6TRANSCRIBE A FILE"N &A 2G9
 %APE 9DICATOR 6TRANSCRIBE A C>RIAGE
 RETURN REPRES5T\$ Z A DEPICT\$ KEY WD
 REQUIRE ! USE (TWO T]M9,N 9DICATORS--"O
 6EXIT , -PUT] ,BRL ,CODE1 ! O!R 6T]M9ATE
 ! %APE 9DICATOR4 ,! F/ T]M9,N 9DICATOR
 WD CLOSE ! %APE 9DICATOR 7! MO/ REC5T
 SYMBOL71 :ILE ! SECOND WD EXIT , -PUT]

,BRL	,DOTS	,M1N+	
"3333	"333333333333	"3333333333333333	
.....		SPACE	
! ""	#BCDF ""	EXCLAM,N P09T	
" ""	#E ""	QUOT,N M>KS1 D\#	
		QUOTES	
# ""	#CDEF ""	NUMB] SIGN1 P.D	
		SIGN 7WEI<T7	
\$ ""	#ABDF ""	DOLL> SIGN	
% ""	#ADF ""	P]C5T SIGN	
	,SYMBOLS3 ,	ASCII ,ORD]	#FI
,BRL	,DOTS	,M1N+	A#EC
"3333	"333333333333	"3333333333333333	
& ""	#ABCDF ""	AMP]S&	
' ""	#C ""	AP0/ROPHE1 S+LE	
		QUOTE	
(""	#ABCEF ""	LEFT P>5!SIS	
) ""	#BCDEF ""	"R P>5!SIS	
* ""	#AF ""	A/]ISK1	
		MULTIPLIC,N SIGN	
-----			#ED
+ ""	#CDF ""	PLUS SIGN	
, ""	#F ""	-MA	
- ""	#CF ""	HYPH51 M9US SIGN	
. ""	#DF ""	P]IOD1 DECIMAL	

```

P09T
/ "" #CD "" SLA%1 DIVI.N SIGN
0 "" #CEF "" #J
1 "" #B "" #A
2 "" #BC "" #B
3 "" #BE "" #C
4 "" #BEF "" #D
5 "" #BF "" #E
6 "" #BCE "" #F
7 "" #BCEF "" #G

,BRL ,DOTS ,M1N+ #EE
"3333 "333333333333 "3333333333333333
8 "" #BCF "" #H
9 "" #CE "" #I
: "" #AEF "" COLON
; "" #EF "" SEMICOLON
< "" #ABF "" LESS ?AN
= "" #ABCDEF "" EQUALS
> "" #CDE "" GRT] ?AN
? "" #ADEF "" "Q M>K
@ "" #D "" AT
A "" #A "" ;,A
B "" #AB "" ;,B
C "" #AD "" ;,C
D "" #ADE "" ;,D
E "" #AE "" ;,E
-----#EF
F "" #ABD "" ;,F
G "" #ABDE "" ;,G
H "" #ABE "" ;,H
I "" #BD "" ;,I
J "" #BDE "" ;,J
K "" #AC "" ;,K
L "" #ABC "" ;,L
,SYMBOLS3 ,ASCII ,ORD] #GA
,BRL ,DOTS ,M1N+ A#EF
"3333 "333333333333 "3333333333333333
M "" #ACD "" ;,M
N "" #ACDE "" ;,N
O "" #ACE "" ;,O
P "" #ABCD "" ;,P
Q "" #ABCDE "" ;,Q
R "" #ABCE "" ;,R
S "" #BCD "" ;,S
-----#EG
T "" #BCDE "" ;,T
U "" #ACF "" ;,U
V "" #ABCF "" ;,V
W "" #BDEF "" ;,W
X "" #ACDF "" ;,X
Y "" #ACDEF "" ;,Y
Z "" #ACEF "" ;,Z
[ "" #BDF "" LEFT BRACKET
\ "" #ABEF "" BACKSLA%
] "" #ABDEF "" "R BRACKET
^ "" #DE "" UP >R[_/C>ET
_ #DEF1 #DEF "USCORE
_@ #DEF1 #D "" GRAVE A35T
-----#EH
A "" #A "" ;,A
,BRL ,DOTS ,M1N+ A#EH
"3333 "333333333333 "3333333333333333
B "" #AB "" ;,B
C "" #AD "" ;,C
D "" #ADE "" ;,D
E "" #AE "" ;,E
F "" #ABD "" ;,F
G "" #ABDE "" ;,G

```



```

H "" #ABE "" ;H
I "" #BD "" ;I
J "" #BDE "" ;J
K "" #AC "" ;K
L "" #ABC "" ;L
M "" #ACD "" ;M
N "" #ACDE "" ;N

```

```
-----#EI
```

```

O "" #ACE "" ;O
P "" #ABCD "" ;P
Q "" #ABCDE "" ;Q
R "" #ABCE "" ;R
S "" #BCD "" ;S
T "" #BCDE "" ;T
U "" #ACF "" ;U
V "" #ABCF "" ;V

```

```

,SYMBOLS3 ,ASCII ,ORD] #GC
,BRL ,DOTS ,M1N+ A#EI
"3333 "333333333333 "3333333333333333
W "" #BDEF "" ;W
X "" #ACDF "" ;X
Y "" #ACDEF "" ;Y
Z "" #ACEF "" ;Z
_[ #DEF1 #BDF LEFT BRACE
_\ #DEF1 #ABEF V]TICAL B>

```

```
-----#FJ
```

```

_] #DEF1 #ABDEF "R BRACE
_^ #DEF1 #DE "" TILDE

```

```

,A4I;NAL ,-PUT] ,BRL ,CODE
,SYMBOLS

```

```

,BRL ,DOTS ,M1N+
"3333 "333333333333 "3333333333333333
_! #DEF1 #BCDF TRANSCRIB]'S OP;N
SYMBOL
_# #DEF1 #DF "" TRANSCRIB]'S OP;N
SYMBOL
_# #DEF1 #CDEF HALF-L9E %IFT UP
_? #DEF1 #ADEF HALF-L9E %IFT D[N
_: #DEF1 #AEF 5D HALF-L9E %IFT
_$ #DEF1 #ABDF 2G9 %APE 9DICATOR
,BRL ,DOTS ,M1N+ A#FJ
"3333 "333333333333 "3333333333333333
_: #DEF1 #AEF 5D %APE 9DICATOR
_+ #DEF1 #CDF 2G9 ,-PUT] ,BRL
,CODE
_: #DEF1 #AEF 5D ,-PUT] ,BRL
,CODE

```

```
-----#FA
```

```

_% #DEF1 #ADF 2G9 ,NEME? ,CODE
_: #DEF1 #AEF 5D ,NEME? ,CODE
_& #DEF1 #ABCDF 3T9U,N 9DICATOR
_( #DEF1 #ABCEF RES]V$
_) #DEF1 #BCDEF RES]V$
_= #DEF1 #ABCDEF RES]V$
_* #DEF1 #AF 2G9 EMPHASIS
_/ #DEF1 #CD 5D EMPHASIS
_ "" #DEF "" %IFT 9DICATOR1
ISOLAT$ L[]-CELL
9DICATOR
_> #DEF1 #CDE CAPS LOCK 9DICATOR
_< #DEF1 #ABF CAPS REL1SE
9DICATOR

```

```

,SYMBOLS3 ,ASCII ,ORD] #GE
,BRL ,DOTS ,M1N+ A#FA
"3333 "333333333333 "3333333333333333

```

```
,NOTE3 ,UPP] CASE LRS MAY 2 TRANSCRIB$ Z
%[N 9 ! TA#1 PREC$$ 0DOTS #DEF1 OR
PREC$$ 0! CAPS LOCK 9Dicator2 APPLICA#
,-PUT] ,BRL ,CODE RULES DET]M9E
TRANSCRIP;N ME?OD4 8,RES]V$0 DESIGNATES
CURR5TLY UNASSIGN$ SYMBOLS : >E RES]V$ =
FUTURE USE4
```

,BRL	,DOTS	,M1N+	
"3333	"3333333333333	"3333333333333333	
.....		SPACE	
A ""	#A ""	,A OR ;A	
B ""	#AB ""	,B OR ;B	
C ""	#AD ""	,C OR ;C	
D ""	#ADE ""	,D OR ;D	
E ""	#AE ""	,E OR ;E	
F ""	#ABD ""	,F OR ;F	
G ""	#ABDE ""	,G OR ;G	
H ""	#ABE ""	,H OR ;H	
I ""	#BD ""	,I OR ;I	
J ""	#BDE ""	,J OR ;J	
	,SYMBOLS3	,BRL ,ORD]	#GG
,BRL	,DOTS	,M1N+	A#FC
"3333	"3333333333333	"3333333333333333	
K ""	#AC ""	,K OR ;K	
-----			#FD
L ""	#ABC ""	,L OR ;L	
M ""	#ACD ""	,M OR ;M	
N ""	#ACDE ""	,N OR ;N	
O ""	#ACE ""	,O OR ;O	
P ""	#ABCD ""	,P OR ;P	
Q ""	#ABCDE ""	,Q OR ;Q	
R ""	#ABCE ""	,R OR ;R	
S ""	#BCD ""	,S OR ;S	
T ""	#BCDE ""	,T OR ;T	
U ""	#ACF ""	,U OR ;U	
V ""	#ABCF ""	,V OR ;V	
X ""	#ACDF ""	,X OR ;X	
Y ""	#ACDEF ""	,Y OR ;Y	
Z ""	#ACEF ""	,Z OR ;Z	
-----			#FE
& ""	#ABCDF ""	AMP]S&	
= ""	#ABCDEF ""	EQUALS	

```

( "" #ABCEF "" LEFT P>5!SIS
! "" #BCDF "" EXCLAM,N P09T
) "" #BCDEF "" "R P>5!SIS

,BRL ,DOTS ,M1N+ A#FE
"3333 "333333333333 "3333333333333333
* "" #AF "" A/]ISK1
MULTIPLIC,N SIGN
< "" #ABF "" LESS ?AN
% "" #ADF "" P]C5T SIGN
? "" #ADEF "" "Q M>K
: "" #AEF "" COLON
$ "" #ABDF "" DOLL> SIGN
] "" #ABDEF "" "R BRACKET
\ "" #ABEF "" BACKSLA%
[ "" #BDF "" LEFT BRACKET
-----#FF
W "" #BDEF "" ;,W OR ;W
1 "" #B "" #A
2 "" #BC "" #B
3 "" #BE "" #C
4 "" #BEF "" #D
5 "" #BF "" #E
6 "" #BCE "" #F
7 "" #BCEF "" #G
8 "" #BCF "" #H
9 "" #CE "" #I
0 "" #CEF "" #J
,SYMBOLS3 ,BRL ,ORD] #GI
,BRL ,DOTS ,M1N+ A#FF
"3333 "333333333333 "3333333333333333
/ "" #CD "" SLA%1 DIVI.N SIGN
+ "" #CDF "" PLUS SIGN
# "" #CDEF "" NUMB] SIGN1 P.D
SIGN 7WEI<T7
-----#FG
> "" #CDE "" GRT] ?AN
' "" #C "" APO/ROPHE1 S+LE
QUOTE
- "" #CF "" HYPH51 M9US SIGN
@ "" #D "" AT
^ "" #DE "" UP >R[_/C>ET
_ "" #DEF "" %IFT 9DICATOR1
ISOLAT$ L[]-CELL
9DICATOR
_A #DEF1 #A "" ;,A
_B #DEF1 #AB "" ;,B
_C #DEF1 #AD "" ;,C
_D #DEF1 #ADE "" ;,D
_E #DEF1 #AE "" ;,E
_F #DEF1 #ABD "" ;,F
_G #DEF1 #ABDE "" ;,G
-----#FH
_H #DEF1 #ABE "" ;,H
,BRL ,DOTS ,M1N+ A#FH
"3333 "333333333333 "3333333333333333
_I #DEF1 #BD "" ;,I
_J #DEF1 #BDE "" ;,J
_K #DEF1 #AC "" ;,K
_L #DEF1 #ABC "" ;,L
_M #DEF1 #ACD "" ;,M
_N #DEF1 #ACDE "" ;,N
_O #DEF1 #ACE "" ;,O
_P #DEF1 #ABCD "" ;,P
_Q #DEF1 #ABCDE "" ;,Q
_R #DEF1 #ABCE "" ;,R
_S #DEF1 #BCD "" ;,S
_T #DEF1 #BCDE "" ;,T
_U #DEF1 #ACF "" ;,U

```

```
-----#FI
_V    #DEF1 #ABCF    ;,V
_X    #DEF1 #ACDF    ;,X
_Y    #DEF1 #ACDEF   ;,Y
_Z    #DEF1 #ACEF    ;,Z
_&    #DEF1 #ABCDF   3T9U,N 9DICATOR
_=    #DEF1 #ABCDEF  RES]V$
```

```
,SYMBOLS3 ,BRL ,ORD]    #HA
,BRL ,DOTS ,M1N+    A#FI
"3333 "333333333333 "3333333333333333
_== 7SPACE7 #DEF1 C.TA# SPACES
      #ABCDEF1    9DICATOR
      #ABCDEF1
      ' ' ' 7SPACE7
_(    #DEF1 #ABCEF  RES]V$
_!    #DEF1 #BCDF   TRANSCRIB]'S OP;N
      SYMBOL
_)    #DEF1 #BCDEF  RES]V$
_*    #DEF1 #AF " " " 2G9 EMPHASIS
_<    #DEF1 #ABF    CAPS REL1SE
      9DICATOR
```

```
-----#GJ
_%    #DEF1 #ADF    2G9 ,NEME? ,CODE
_?    #DEF1 #ADEF   HALF-L9E %IFT D[N
_:    #DEF1 #AEF    T]M9,N 9DICATOR
_:    #DEF1 #AEF    5D , -PUT] ,BRL
      ,CODE
_:    #DEF1 #AEF    5D ,NEME? ,CODE
_:    #DEF1 #AEF    5D %APE 9DICATOR
_:    #DEF1 #AEF    5D HALF-L9E %IFT
_$    #DEF1 #ABDF   2G9 %APE 9DICATOR
_]    #DEF1 #ABDEF   "R BRACE
_\    #DEF1 #ABEF   V]TICAL B>
,BRL ,DOTS ,M1N+    A#GJ
"3333 "333333333333 "3333333333333333
_[    #DEF1 #BDF    LEFT BRACE
_W    #DEF1 #BDEF   ;,W
_/    #DEF1 #CD " " " 5D EMPHASIS
```

```
-----#GA
_+    #DEF1 #CDF    2G9 , -PUT] ,BRL
      ,CODE
_#    #DEF1 #CDEF   HALF-L9E %IFT UP
_>    #DEF1 #CDE    CAPS LOCK 9DICATOR
_@    #DEF1 #D " " " " GRAVE A35T
_^    #DEF1 #DE " " " " TILDE
_    #DEF1 #DEF     "USCORE
_ .    #DEF1 #DF " " " " TRANSCRIB]'S OP;N
      SYMBOL
. " " " " #DF " " " " " " " " P]IOD1 DECIMAL
      P09T
" " " " #E " " " " " " " " QUOT,N M>KS1 D\#
      QUOTES
; " " " #EF " " " " " " " " SEMICOLON
, " " " #F " " " " " " " " -MA
```

```
,SYMBOLS3 ,BRL ,ORD]    #HC
```

```

,,9DEX                                #GB-GC

,A2REVI,NS1 #G
,ACRONYMS1 #G1 #AH
,A4I;NAL ,,CBC SYMBOLS1 #FJ-FA
,AMBIGUITIES1 CL>IFIC,N (1 #G1 #CG1 #CH
,>R[3
  ,D[NW>D-CURVE1 #BA
  ,"R-PO9T+1 #BF
  ,UP1 #G1 #DG
,,ASCI11 DEF9I;N1 #G
,,ASCII ,ORD]1 SYMBOLS1 #EC-FA

,BLANK L9ES1 #AA1 #AG
,BRL ,ORD]1 SYMBOLS1 #FC-GA

8,C>RIAGE ,RETURN01 #BA
,CAPS LOCK1 #AG1 #BJ1 #CG
  ,TRANSCRIB]'S NOTE1 #AI
,CAPS REL1SE1 #BJ
,CASE DESIGN,N1 #G1 #AI-BJ1 #EC1 #FC
,-PUT] ,BRL ,CODE ,SYMBOLS3
  ,,,ASCII ,ORD]1 #EC-FA
  ,BRL ,ORD]1 #FC-GA
,3T9U,N 9DICATOR1 #BC-BD                #HE
,3TRAC;NS1 #G1 #I                        A#GC
,3TROL "*"S1 #BA
,C.TA# SPACES1 #BC1 #BI-CA
  ,DIVID+ L9ES1 #BC1 #CJ-CA
  ,9DICATOR1 #BC1 #BI-CA

,DEVIS+ SYMBOLS1 #AA1 #DC1 #DD1 #DE
,DIVID+ L9ES1 #AC1 #BC-BD1 #CJ-CA

;,E-MAIL A4RESSES1 #F1 #BJ1 #CG
,EMB$D$ NOT,N1 #H1 #AG-AH1 #EA
  ,=MAT1 #H1 #AG-AH1 #EA
  ,RUNOV]S1 #AG-AH1 #BC
,EMPHASIS 9DICATORS1 #AG1 #CC-CF1 #CG

,FOOTNOTE 9DICATORS1 #DI
,=MAT1 #AA
  ,EMB$D$ NOT,N1 #H1 #AG-AH1 #EA
  ,PROGRAM L9ES1 #I1 #AA-AF1 #AI-BA1
    #BC1 #BE-BG1 #BI-CA
,FRAC;N L9E1 HORIZONTAL1 #DA
,FREQU5T SWIT*ES1 #CG1 #CH

,HALF-L9E %IFTS1 #I1 #DG-DH1 #EA
  ,T]M9AT$ 0SPACE1 #I1 #DH
,HORIZONTAL & V]TICAL SPAC+1 #G        B#GC
,HORIZONTAL FRAC;N L9E1 #DA

```

```
,ID5TIFY+1 #A-F
,-;TS EXPLA9+1 #C-D
,DOT MATRIX PR9T1 #B
,5CLOSURE SYMBOLS1 #D
,HORIZONTAL SIGNS ( -P>ISON1 #E
,PROGRAM L9ES1 #A-C
,SYMBOL N 9 PRIM>Y CODE1 #F
,TYPE /YLES1 #A
,^WS \ ( 3TEXT1 #F
,9D5T$ MAT]IAL1 #C1 #BE-BF
,9DICATORS3
,-PUT] ,BRL ,CODE1 #AG1 #CG
,CAPS LOCK1 #AG1 #BJ1 #CG
,CAPS REL1SE1 #BJ
,CASE1 #AI-BJ
,3T9U,N1 #BC-BD
,C.TA# SPACE1 #BC1 #BI-CA
,EMPHASIS1 #AG1 #CC-CF1 #CG
,NEME? ,CODE1 #AG1 #CG1 #CI-DA1 #DI1
#EA
,REF];E1 #DI
,RES]V$1 #EC1 #FC #HG
,%APE1 #E1 #AG1 #DC-DD1 #EA C#GC
,%IFT1 CASE1 #AI-BJ
,%IFT1 HALF-L9E1 #DG-DH1 #EA
,T]M9,N1 #AG1 #DC1 #DG1 #EA
,ISOLAT$ L[]-CELL SIGNS1 #BG

,KEYS1 DEPICT$1 #BA1 #DC-DD1 #EA
-----#GD
,LABEL = PROGRAM1 #AC-AF
,L[]-CELL SIGNS1 ISOLAT$1 #BG

,NEME? ,CODE 9DICATORS1 #AG1 #CG1 #CI-
#DA1 #DI1 #EA
,NUMB]S1 #G

,PRIM>Y CODE1 #G1 #AA1 #DI1 #EA
,PR9TA# SYMBOLS1 #G
,PROGRAM L9ES1 #A-F1 #I1 #AA-AB1 #AC1
#AD-AF1 #AI-BA1 #BC1 #BE-BG1 #BI-CA1
#CD1 #DC1 #DE1 #DI
,PROGRAMS1 #A
,CASE1 #AI
,-PLETE ON PAGE1 #AC
,DIVID+ L9ES1 #AC
,EXAMPLE1 #B1 #C1 #I1 #AB1 #AD-AF1
#AI1 #BE-BF1 #BI1 #CJ1 #CA1 A#GD
#CD1 #DI
,=MAT1 #B1 #C1 #I1 #AA-AB1 #AC1 #AD-
#AF1 #BE-BF
,LABEL1 #AC-AF
,RUNOV]S1 #AA-AF1 #BE1 #BF1 #CJ1 #CA
,PUNCTU,N1 #G1 #EA

,REF];E 9DICATORS1 FOOTNOTES1 #DI
,RES]V$ 9DICATORS1 #EC1 #FC
,"R-PO9T+ >R[ SYMBOL1 #BF
,RUNOV]S1 #BC
,C.TA# SPACES1 #CJ-CA
,EMB$D$1 #AG-AH1 #BC
,9D5T$ MAT]IAL1 #BE1 #BF
,PROGRAM L9ES1 #AA-AF1 #BE1 #BF1 #CJ1
#CA

,SET-(F MAT]IAL1 #AA-AF1 #BC1 #CH
,%APE 9DICATORS1 #E1 #AG1 #DC-DD1 #EA
,TRANSCRIB]'S NOTE1 #DC1 #DD
,%IFT3
```

```

,5D HALF-L9E1 #DG-DH1 #EA
,HALF-L9E D[N1 #I1 #DG-DH
,HALF-L9E UP1 #DG #HI
,%IFT 7CASE7 9Dicator1 #AI-BJ B#GD
,SPACES1 C.TA#1 #BC1 #BI-CA
,RUNOV]S1 #BC1 #CJ-CA
,SPACES1 SIGNIFICANT1 #BJ1 #BI-CA
,SPAC+1 V]TICAL & HORIZONTAL1 #G
,SPECIAL SYMBOLS PAGE1 #AA
,SUBSCRIPTS1 #I1 #DG-DI
,SUP]IMPOS$ SYMBOLS1 #E
,SUP]SCRIPTS1 #DG1 #DI
,SY/EMATIC 9D5T,N1 #C1 #BE-BF
,SYMBOLS3
,,ASCII ORD]1 #EC-FA
,BRL ORD]1 #FC-GA
,DEVIS$1 #AA1 #DC1 #DD1 #DE
,PR9TA#1 #G
,SUB/ITU;NS1 #DE
,SUP]IMPOS$1 #E

,T]M9,N 9Dicator1 #AG1 #DC1 #DG1 #EA
,TO1 96& BY1 #G1 #I
,TRANSCRIB]'S NOTES1 #AA
,CASE1 #AI
,PR9T =MAT *ANGE1 #DJ
,%APE 9Dicators1 #CD1 #DC1 #DD
,SUB/ITU;NS1 #DE
,TRANSCRIB]'S OP;N SYMBOL1 #CD1 C#GD
#CE1 #DE
,TYPE FONTS1 #CC-CE
,TRANSCRIB]'S OP;N SYMBOLS1 #CD1 #CE1
#DE
,TYPE FONTS1 #CC-CE

,V]TICAL & HORIZONTAL SPAC+1 #G

,WEB ,SITE A4RESSES1 #H1 #I1 #AC1 #AH1
#BD

```

#IA

